EDUCATION WING
EMBASSY OF INDIA
KATHMANDU, NEPAL
BE. Model Entrance Examination

Attempt all Questions. Each question carries one mark.

Choose the correct answer and blacken the appropriate bubble using HB pencil on answer sheet.

SET- I

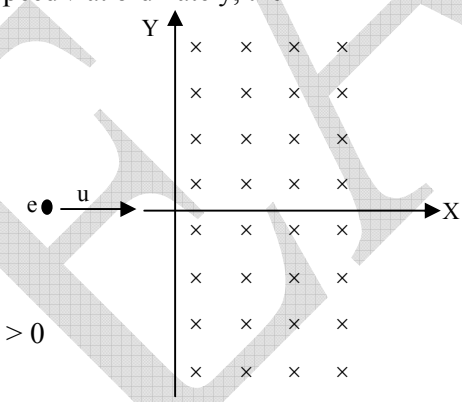
Subject: Physics

Time: 1 hr

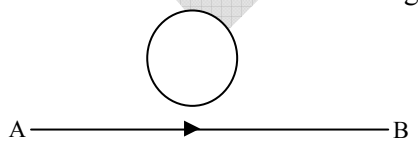
1. If $P_x = 2\sin t$ and $P_y = 2\cos t$, then the angle between linear momentum **p** and force **F** will be
   a. 30^0 b. 45^0 c. 60^0 d. 90^0
2. The displacement x of a particle varies with time as
   $$x = ae^{-\alpha t} + be^{\beta t}$$
   Where a, b, α and β are positive constants.
   The velocity of the particle will
   a. go on decreasing with time
   b. go on increasing with time
   c. be independent of α and β
   d. drop to zero where $\alpha = \beta$
3. A projectile is fired from level ground at an angle θ above the horizontal. The elevation angle ϕ of the highest point as seen from the launch point is related to θ by the relation
   a. $\tan\phi = \frac{1}{4}\tan\theta$ b. $\tan\phi = \tan\theta$ c. $\tan\phi = \frac{1}{2}\tan\theta$ d. $\tan\phi = 2\tan\theta$
4. A particle of mass m is moving in a horizontal circle of radius r under a centripetal force equal to $-k/r^2$, where k is a constant. The total energy of the particle is
   a. $\frac{k}{r}$ b. $\frac{k}{2r}$ c. $-\frac{k}{2r}$ d. $\frac{k}{r^2}$
5. An object is kept on a smooth inclined plane of 1 in *l*. The horizontal acceleration to be imparted to the inclined plane so that the object is stationary relative to the incline is given by
   a. $g\sqrt{(l^2 - 1)}$ b. $g(l^2 - 1)$ c. $g / \sqrt{(l^2 - 1)}$ d. $g / (l^2 - 1)$
6. A solid sphere and a hollow sphere of the same material and of a same size can be distinguished without weighing
   a. by determining their moments of inertia about their coaxial axes
   b. by rolling them simultaneously on an inclined
   c. by rotating them about a common axis of rotation
   d. by applying equal torque on them
7. A bird, resting on the floor of an air tight box which is being carried by a boy, starts flying. The boy will feel that the box is now
   a. heavier
   b. lighter
   c. shows no change in weight
   d. lighter in the beginning and heavier afterwards

8. An elastic string has a length a when tension in it is 4 N. Its length is b when tension is 5 N. On subjecting the string to a tension of 9 N its length will be
   a. a + b b. b – a c. (5b – 4a) d. [(a + b / (b – a))]
9. An ideal heat engine working between temperature T and T′ has efficiency η. If both the temperatures are raised by 100K each, the new efficiency of the heat engine will be
   a. equal to η
   b. greater than η
   c. less than η
   d. greater or less than η depending upon the nature of the working substances
10. An ideal gas at pressure P is adiabatically compressed so that its density becomes n times the initial value. The final pressure of the gas will be
   a. n^{γ} P b. $n^{-\gamma}$ P c. $n^{(\gamma-1)}$ P d. $n^{(1-\gamma)}$ P
11. A mono atomic gas supplied the heat Q very slowly keeping the pressure constant. The work done by the gas will be
   a. $\frac{2}{3}$Q b. $\frac{3}{5}$Q c. $\frac{2}{5}$Q d. $\frac{1}{5}$Q
12. The Wien's displacement law expresses relation between
   a. wavelength corresponding to maximum energy and temperature
   b. radiation energy and wavelength
   c. temperature and wavelength
   d. colour of light and temperature
13. The pressure exerted by an electromagnetic wave of intensity I (watt/m^2) on a nonreflecting surface is (c is the velocity of light)
   a. Ic b. Ic^2 c. I/c d. I/c^2
14. If white light is used in Young's double-slit experiment, then a very large number of coloured fringes can be seen
   a. with a central black fringe
   b. with a central white fringe
   c. first order red fringes being closer to the central white fringe
   d. with first order violet fringes being closer to the central while fringe
15. The angle of the prism for which there is no emergent ray will be, if its critical angle is i_c
   a. $2i_c$ b. less than i_c c. $2i_c$ d. greater than $2i_c$
16. The velocity of sound in air is *v* and the root mean square velocity of the molecules is c. Then *v*/c is equal to
   a. γ/3 b. $(\gamma/3)^{1/2}$ c. $\gamma/\sqrt{3}$ d. $\sqrt{\gamma}/3$
17. The fork A of frequency 100 is sounded with another tuning fork B. The number of beats produced is 2. On putting some wax on the prong of B, the number of beats reduces to 1. The frequency of the fork B is
   a. 101 b. 99 c. 102 d. 98
18. The length of each side of a cubical closed surface is *l*. If charge q is situated on one of the vertices of the cube, then the flux passing through each face of the cube will be
   a. $\frac{q}{6\varepsilon_0}$ b. $\frac{q}{12\varepsilon_0}$ c. $\frac{q}{24\varepsilon_0}$ d. $\frac{\varepsilon_0}{24q}$

19. If a positive charge is shifted from a low potential region to a high potential region, the electric potential energy
    a. decreases b. increases
    c. remains the same d. may increase or decrease
20. Three similar cells, each of e.m.f 2V and internal resistance rΩ send the same current through and external resistance 2Ω. When connected in series or in parallel. The strength of the current flowing through the external resistance is
    a. 2A b. 0.75A c. 1A d. 1.5A
21. Two cells of same e.m.f are connected in series. Their internal resistance are r_1 and r_2 respectively and $r_1 > r_2$. When this combination is connected to an external resistance R, then the potential difference between the terminals of first cell becomes zero. In this condition, the value of R will be
    a. $r_1 + r_2$ b. $r_1 - r_2$ c. $(r_1 + r_2)/2$ d. $(r_1 - r_2)/2$
22. An electron with a speed u along the positive x-axis at y = 0 enters a region of uniform magnetic field $\mathbf{B} = -B_0\mathbf{k}$ which exists to the right of y-axis. The electron exists from the region after some time with the speed *v* at ordinate y, then



    a. $v > u, y < o$ b. $v = u, y > 0$
    c. $v > u, y > 0$ d. $v = u, y < 0$
23. A magnetic needle is kept in a non-uniform magnetic field. It experiences
    a. a torque but not a force b. neither a force nor a torque
    c. a force and a torque d. a force but not a torque
24. The current flows from A to B as shown in fig. The direction of the induced current in the loop is



    a. clockwise b. anticlockwise c. straight line d. none of these
25. In an L-R circuit, time constant is that time in which current grows from zero to the value
    a. $0.63I_0$ b. $0.50I_0$ c. $0.37I_0$ d. I_0
26. When a certain metallic surface is illuminated with monochromatic light of wavelength λ, the stopping potential for photoelectric current is $3V_0$. When the same surface is illuminated with light of wavelength 2λ, the stopping potential is V_0. The threshold wavelength for this surface for photoelectric effect is
    a. 6λ b. 4λ/3 c. 4λ d. 8λ

27. How many times does the electron go around the first Bohr orbit on one second?
a. $\frac{h}{4\pi^2 mr^2}$ b. $\frac{h}{4\pi mr}$ c. $\frac{h}{2\pi mr^2}$ d. $\frac{4\pi mr^2}{h}$
28. The activity of a radioactive substance at t_1 is A_1 and at time t_2 is A_2. If the mean life of the substance is T_m, then the number of atoms decayed during time interval $(t_2 - t_1)$ will be
a. $A_1t_1 - A_2t_2$ b. $(A_1 - A_2)$ c. $(A_1 - A_2)/T_m$ d. $(A_1 - A_2)T_m$
29. In a semiconducting material the mobilities of electrons and holes are μ_e and μ_n respectively. Which of the following is true?
a. $\mu_e > \mu_n$ b. $\mu_e < \mu_n$ c. $\mu_e = \mu_n$ d. $\mu_e < 0; \mu_n > 0$
30. At what velocity the mass of a particle is equal to twice rest mass?
a. c b. c/2 c. 2c/3 d. $\sqrt{3}$ c/2

Subject: Chemistry

Time: 1 hr

1. The mass percentage of carbon in carbon dioxide is
a. 3.4% b. 27.27% c. 34% d. 38.42%
2. The number of moles of $KMnO_4$ required to oxidize 1 mole of FeC_2O_4 completely in acidic medium will be
a. 0.6 b. 0.4 c. 7.5 d. 0.2
3. The total number of nodes in 4f orbital is
a. 4 b. 3 c. 2 d. 1
4. In the nuclear rxn, $x \xrightarrow{-\alpha} y \xrightarrow{-2\beta} z$. x and z are
a. isotopes b. isotones c. isobars d. isodiaphers
5. Which one of the following ions will be the smallest in size?
a. Na^+ b. Mg^{++} c. F^- d. O^{--}
6. In PF_5, the hybridization of P is
a. sp^3 d. dsp^2 c. sp^3d d. sp^3d^2
7. All gases occupy zero volumes when the temperature is reduced to
a. 273^0C b. 273k c. -273^0C d. 0^0C
8. For a cyclic process, which of the following is not true?
a. $\Delta H = 0$ b. $\Delta E = 0$ c. $\Delta G = 0$ d. $W = 0$
9. For the reaction, A + B ⇌ C. The unit of equilibrium constant is
a. litre mol^{-1} b. mole litre c. mole $litre^{-1}$ d. no unit
10. The value of x and y in the redox reaction are $xCl_2 + 6OH^- \rightarrow ClO_3^- + yCl^- + 3H_2O$
a. x = 5, y = 3 b. x = 3, y = 5 c. x = 2, y = 4 d. x = 4, y = 2
11. The volume strength of 1.5 NH_2O_2 solution is
a. 4.8 b. 5.2 c. 8.4 d. 9.4
12. Which of the following alkali metal ions has lowest mobility in aqueous solution?
a. Rb^+ b. Cs^+ c. Li^+ d. Na^+
13. Lead pencil contains
a. Pb b. PbS c. PbO d. graphite
14. Kjeldahl's method is used in the estimation of
a. Nitrogen b. Halogen c. Sulphur d. Oxygen
15. Which of the following compound with formula C_5H_{10} yields acetone on ozonolysis?
a. 3–methyl–1–butene b. 2–methyl–1–butene
c. Cyclopentane d. 2–methyl–2–butene

16. Which gas is responsible for 'Bhopal gas trgedy' in 1984?
    a. CO b. Methyl isocyanate
    c. SO_2 and NO_2 d. Ethyl isocyanate
17. The number of atoms in a face centred cubic unit cell is
    a. 4 b. 6 c. 8 d. 12
18. The freezing point of a solution prepared from 1.25 g of non-electrolyte and 20g of water is 271.94k. If k_f is 1.86k kg mol^{-1}. Then the molar mass of solute is.
    a. 109 b. 106 c. 115 d. 93

19. On passing C ampere of electricity through an electrolyte solution. The equivalent weight E of the metal is
    a. $E = \frac{c \times t}{m \times 96500}$ b. $E = \frac{c \times m}{t \times 96500}$
    c. $E = \frac{96500 \times m}{c \times t}$ d. $E = \frac{c \times t \times 96500}{m}$
20. A first order reaction is 87.5% completed in 30 minutes. The half-life of the reaction is.
    a. 10 minutes b. 15 minutes c. 20 minutes d. 25 minutes
21. Small liquid droplets dispersed in another liquid is called.
    a. gel b. suspension c. emulsion d. true solution
22. Which of the following is nitric acid anhydride?
    a. NO b. NO_2 c. N_2O_3 d. N_2O_3
23. Which of these metal is not present in German silver?
    a. Cu b. Ni c. Zn d. Sn
24. The effective atomic number of copper in $[Cu\ (CN_4]^2$ ion is
    a. 31 b. 33 c. 35 d. 36
25. Chloroform on treatment with conc. HNO_3 gives.
    a. Chloropicrin b. Nitromethane c. Picricacid d. Acetylene
26. Which of the following compounds is oxidized to prepare the methyl ethyl ketone?
    a. 2–propanol b. 2–Butanol c. 1–Butanol d. 1–Butyl alcohol
27. Benzaldehyde reacts with ethanoic KCN to give
    a. $C_6H_5CHOHCN$ b. C_6H_5 CHOHCOOH
    c. $C_6H_5CHOHCOC_6H_5$ d. $C_6H_5CHOHCHOHC_6H_5$
28. Reaction $CH_3CONH_2 \xrightarrow{NaOBr}$ gives
    a. CH_3Br b. CH_4 d. CH_3COBr d. CH_3NH_2
29. Which base is present in RNA but not in DNA?
    a. Guanine b. Cytosine c. Uracil d. Thymine
30. Caprolactam polymerises to give
    a. Terylene b. Teflon c. Glyptal d. Nylon–6

Subject: English

Time: 1 hr.

Read the passage and answer the given question (1 to 5)

The great Acharyas have said that having discovered a great goal, surrender yourself to that goal and act towards it, drawing your inspiration from that goal, whereby you will get a new column of energy. Do not allow this energy to be dissipated in the futile memories of past regrets of failure, or in the excitement of the present, and thus bring that entire energy focused into

activity. That is the highest creative action in the world outside. Thereby, the individual who is till now considered most inefficient finds his way to the highest achievement and success.

This is said very easily in a second. But in order to change the mind to this attitude needs considerable training because we have already trained the mind wrongly to such an extent that we have become perfect in imperfections.

Not knowing the art of action, we have been master artists in doing the wrong thing. The totality of activity will bring the country to a wrong end, indeed.

If each one is given a car, to achieve an ideal socialistic pattern, and nobody knows driving but everybody starts driving, what would be the condition on the road? Everybody has equal right on the public road. Then each car must necessarily dash against the other, and there is bound to be a jumble. This seems to be the very apt pattern of life that we are heading to. Everyone of us is vehicle. We know how to go forward. The point is, intellect is very powerful and everyone is driving but nobody seems to know how to control the mental energy and direct it properly, or guide it to the proper destination.

1. Which of the following is the source of energy?
   a. A column that supports a building
   b. Stimulation obtained from a set aim.
   c. Highest creative action
   d. Proper training of the mind to achieve perfection
2. The author's chief concern is
   a. establishment of socialistic b. struggle for equal rights
   c. discovery of a great goal in life d. regulation of energy in the proper direction
3. Which of the following will cause the country to perish?
   a. Directing mental energy to the right destination
   b. Driving cars without proper driving knowledge and skill
   c. Wrong deeds performed without proper knowledge
   d. Memories of past regrets and failures
4. Which of the following could lead to success?
   a. Cherishing the memories of the past
   b. Being alert about the excitement of the present
   c. Bringing all the energy into activity
   d. None of these
5. What is the effect of training of the wrong mind?
   a. We have become perfect in all aspects b. Art of action is too much emphasized
   c. Each of us could become a master artist d. None of these
6. The medal was bestowed _____the soldier
   a. on b. at c. to d. n
7. We went _____
   a. on a picnic to Vedatar in spring b. to Vedatar on a picnic in spring
   c. on a picnic to Vedatar in spring d. to a picnic to Vedatar in spring
8. Retrospective means ____
   a. to look back b. to look ahead c. to detest d. to move side wards
9. Getting a good start in life _____to a few in life
   a. happen b. happens c. were happening d. have happened
10. While everyone else _____, she_____ quietly in the kitchen
   a. laughed, cried b. was laughing, was crying
   c. laughed, was crying d. laughed, was cried

11. _____ want to know the facts
    a. I , his mother and you b. His mother, you and I
    c. You, his mother and I d. I, you and his mother
12. The couple got married _____the 23 of June, 1987
    a. by b. in c. on d. from
13. The woman has the capacity ____cooking for a large group
    a. with b. for c. for d. of
14. A route to be followed on a journey is _____
    a. itinerary b. trek c. wander d. cartography
15. Something existing only in name _____
    a. pseudonym b. anonymous c. pen name d. titular

Choose the right analogies:

16. Night is to Nocturnal in the same way as Vernal is to _____
    a. winter b. summer c. autumn d. spring
17. Restaurant: Menu
    a. library : catalogue b. journal : newspaper
    c. book : encyclopedia d. college : account
18. Clanking : steel
    a. hoofs : clattering b. rain : pattering
    c. ripping : paper d. water : flowing

Choose the right alternative

19. He said to her, "Don't read so fast."
    a. He told her not to read so fast. b. He advised her not to read so fast.
    c. He requested her not to read so fast d. He asked her not to raced so fast
20. Gerontology deals with _____
    a. you juvenile children b. old aged people
    c. unborn babies d. middle aged persons

Choose the right synonym (Q. 21-24)

21. Beget
    a. To be the son of b. To be the father of
    c. To be the daughter of d. To a friend of
22. Diligent
    a. callous b. laxity c. industrious d. decrepit
23. Humane
    a. charitable b. violent c. uncivilized d. merciless
24. Gratis
    a. free b. salaried c. compensate d. sinecure

Choose the right opposite word (25-26)

25. Hideous
    a. ugly b. shocking c. dreadful d. lovely
26. Humdrum
    a. mundane b. lively c. serene d. ordinary

Choose the right alternative (Q.27-30)

27. _____ enough time, He would have attended the meeting
    a. Had he had b. He would have had
    c. Would he have had d. He had would have

28. She must attend the meeting, _____?
    a. mustn't she b. need she c. could she d. needn't she
29. The _____are responsible for making the decision
    a. committee b. police c. audience d. congress
30. He is _____doing the job well
    a. intent to b. intent at c. intent on d. intent with

Subject: Mathematics

Time: 1 hr

1. The value of $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right)$ is
    a. -1 b. 0 c. $-i$ d. i
2. The number of solutions of $\log_4(x-1) = \log_2(x-3)$:
    a. 3 b. 1 c. 2 d. 0
3. If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, then $\alpha =$
    a. ± 3 b. ± 2 c. ± 5 d. 0
4. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, then the value of $C_0 + 2C_1 + 3C_2 + \ldots (n+1)C_n$ will be
    a. $(n+2)2^{n-1}$ b. $(n+1)2^n$ c. $(n+1)2^{n-1}$ d. $(n+2)2^n$
5. If $a^x = b^y = c^z$ and a, b, c are in G.P. then x, y, z are in
    a. A.P. b. G.P. c. H.P. d. Only equal numbers
6. Period of the function $f(x) = 4\cos\frac{x}{4} + 5\sin\frac{x}{5}$ is
    a. 40π b. π c. 2π d. 20π
7. The domain of the function $2^x + 2^y = 2$ is
    a. (0, 1] b. [0, 1] c. $(-\infty, 0)$ d. $(-\infty, 1)$
8. If f(x) = cosx cos2x cos4x cos8x cos16x, then $f'\left(\frac{\pi}{4}\right)$ is
    a. $\sqrt{2}$ b. $\frac{1}{\sqrt{2}}$ c. 1 d. $\frac{\sqrt{3}}{2}$
9. The equation of tangent to the curve $\frac{x^2}{3} - \frac{y^2}{2} = 1$ which is parallel to y = x is
    a. $y = x \pm 1$ b. $y = x - \frac{1}{2}$ c. $y = x + \frac{1}{2}$ d. $y = 1 - x$
10. $\int 2^{2^{2^x}} 2^{2^x} 2^x \, dx$ is
    a. $\frac{2^{2^x}}{(\ln 2)^3} + c$ b. $\frac{2^{2^{2^x}}}{(\ln 2)^3} + c$
    c. $2^{2^{2^x}}(\ln 2)^3 + c$ d. $2^{2^x} + \frac{x}{2} + c$

11. In a G.P. of positive terms, any term is equal to the sum of the next two terms. The common ratio of G.P. is

a. $2\cos 18^0$ b. $\sin 18^0$ c. $\cos 18^0$ d. $2\sin 18^0$

12. $\int_0^{2\pi} \frac{dx}{e^{\sin x}+1}$ is

a. π b. 2π c. $\frac{\pi}{2}$ d. $\frac{3\pi}{2}$

13. The area bounded by the curve y = tanx and y = cotx in the interval $\left[0, \frac{\pi}{2}\right]$ is

a. $\log\sqrt{2}$ b. log2 c. $\frac{1}{2}\log\sqrt{2}$ d. $\frac{1}{\log 2}$

14. Distance between the pair of lines represented by the equation $x^2 - 6xy + 9y^2 + 3x - 9y - 4 = 0$

a. $\frac{15}{\sqrt{10}}$ b. $\frac{1}{2}$ c. $\sqrt{\frac{5}{2}}$ d. $\frac{1}{\sqrt{10}}$

15. The number of common tangents to the circles
$x^2 + y^2 + 2x + 8y - 23 = 0$ and $x^2 + y^2 - 4x - 10y + 9 = 0$ is

a. 1 b. 2 c. 3 d. 4

16. The eccentricity of the hyperbola conjugate to $x^2 - 3y^2 = 2x + 8$ is:

a. $\frac{2}{\sqrt{3}}$ b. $\sqrt{3}$ c. 2 d. 4

17. For any three non-collinear vectors $\vec{a}, \vec{b}, \vec{c}$ $\vec{a} + 2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 3\vec{c}$ is collinear with $\vec{a}$

What is the value of $\vec{a} + 2\vec{b} + 6\vec{c}$?

a. 0 b. $3\vec{a} + 2\vec{b} + \vec{c}$ c. $4\vec{a} + 3\vec{b} + 2\vec{c}$ d. $5\vec{a} + 3\vec{b} - \vec{c}$

18. If $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = \frac{1}{e}$, $-\pi < \alpha, \beta < \pi$, then total number of ordered pair of (α, β) is

a. 0 b. 1 c. 2 d. 4

19. The shadow of a tower is found to be 60 m shorter when the sun's altitude changes from 30^0 to 60^0. The height of the tower from the ground is approximately equal to:

a. 62 m b. 301 m c. 52 m c. 75 m

20. In a throw of a die the probability of getting one in even number of throw is

a. $\frac{5}{36}$ b. $\frac{5}{11}$ c. $\frac{6}{11}$ d. $\frac{1}{6}$

21. A copper sphere of radius 3 cm is beaten and drawn into a wire of diameter 0.2 cm. The length (in metres) of the wire is:

a. 9 b. 12 c. 18 d. 36

22. In a family of 5 members average age at present is 23 yrs. The youngest member is 7 yrs old. The average age of the family at the birth of youngest member was

a. 20 yrs b. 18 yrs c. 21 yrs d. 22 yrs

23. The present worth of a bill due at the end of 4 yrs is Rs. 1150 and if the bill were due at the end $2\frac{1}{2}$ yrs, it present worth would be Rs. 1240. Find the rate percent.

a. 6% b. 7% c. 8% d. 9%

24. Goods are sold so that when 4% is taken off the sale price, a profit of 20% is made. How much percent is the sale price higher than the cost price?

a. 20% b. 25% c. 30% d. $33\frac{1}{3}$ %

25. A is thrice as good a workman as B and is therefore able to finish a piece of work in 60 days less than B. Find the time in which they can do it working together.

a. $22\frac{1}{2}$ days b. $25\frac{1}{2}$ days c. $27\frac{1}{2}$ days d. $28\frac{1}{2}$ days

26. At an election where there are two candidates only, the candidate who gets 43% of the votes is rejected by a margin of 420 votes. The total number of votes polled were:

a. 3000 b. 2500 c. 1500 d. 3500

27. If sets A and B are defined as $A = \{(x, y) : y = e^x, x \in R\}$ and $B = \{(x, y) : y = x, x \in R\}$ then

a. $B \subset A$ b. $A \subset B$ c. $A \cap B = \phi$ d. $A \cap B = A$

28. A is a working and B a sleeping partner in a business. A puts in Rs. 5000 and B puts in Rs. 6000. A receives $12\frac{1}{2}$ % of the profits for managing the business and the rest is divided in proportion to their capitals.

What does each gets out of a profit of Rs. 880?

a. A : 460, B : 410 b. A : 460, B : 420

c. A : 410, B : 400 d. A : 350, B : 435

29. If $x + \frac{1}{x} = 3$, the value of $x^8 + \frac{1}{x^8}$ will be equal to

a. 2207 b. 47 c. 7 d. 2209

30. If $\sin(\cot^{-1}(x + 1)) = \cos(\tan^{-1}x)$, then x =

a. $-\frac{1}{2}$ b. $\frac{1}{2}$ c. 0 d. $\frac{9}{4}$

….The End….

Answers: Set I

Subject: Physics

1.d	2.b	3.c	4.c	5.c	6.b	7.c	8.c	9.c	10. a
11. c	12. a	13. c	14. b	15. d	16. b	17. c	18. c	19. b	20. b
21. b	22. d	23. c	24. a	25. a	26. c	27. a	28. d	29. a	30. d

Subject: Chemistry

1.b	2.a	3.b	4.a	5.b	6.c	7.c	8.d	9.a	10. c
11. c	12. c	13. d	14. a	15. d	16. b	17. a	18. a	19. c	20. a
21. c	22. d	23. d	24. c	25. a	26. b	27. c	28. d	29. c	30. d

Subject: English

1.b	2.d	3.c	4.c	5.d	6.a	7.b	8.a	9.b	10. b
11. c	12. c	13. b	14. a	15. d	16. d	17. a	18. c	19. a	20. b
21. b	22. c	23. a	24. a	25. d	26. b	27. a	28. d	29. b	30. c

Subject: Mathematics

1.d	2.b	3.a	4.a	5.c	6.a	7.d	8.a	9.a	10. b
11. d	12. a	13. b	14. c	15. b	16. c	17. a	18. d	19. c	20. b
21. d	22. a	23. a	24. b	25. a	26. a	27. c	28. b	29. a	30. a

PEA

EDUCATION WING
EMBASSY OF INDIA
KATHMANDU, NEPAL
BE. Model Entrance Examination

Attempt all Questions. Each question carries one mark.
Choose the correct answer and blacken the appropriate bubble using HB pencil on answer sheet.

SET- II

Subject: Physics

Time: 1 hr.

1. Which of the following is not a unit of Young's modulus?
   a. Nm^{-1} b. Nm^{-2} c. dyne Cn^{-2} d. mega pascal
2. Two bullets are fired horizontally with different velocities from the same height which will reach the ground first?
   a. slower one b. faster one
   c. both will reach simultaneously d. it can't be predicted
3. Which of the following is largest, when the height attained by the projectile is the largest?
   a. range b. time of flight
   c. angle of projectile with vertical d. none
4. An ice cube contains a large air bubble. The cube is floating on the surface of water contained in a trough. What will happen to the water level, when the cube melts?
   a. it will remains unchanged b. it will fall
   c. it will rise d. it will fall and then rise
5. In gravity free space the liquid in a capillary tube will rise to
   a. same height as an earth
   b. less height as on earth
   c. slightly more height than that on earth
   d. infinite height
6. If frequency of the particle executing SHM is 'x' then the frequency of the KE is?
   a. 4x b. x c. 2x d. x/2
7. An iron ball is heated. The percentage increase will be largest in
   a. diameter b. surface area c. volume d. density
8. If the atmospheric temperature and the dew point are nearly equal
   a. relative humidity is nearly 100% b. relative humidity is zero
   c. relative humidity is 50% d. nothing can be said
9. The number of translational degrees of freedom for a diameter gas is
   a. 2 b. 3 c. 5 d. 6
10. The angle between particle velocity and wave velocity in a transverse wave is
    a. zero b. $\pi/4$ c. $\pi/2$ d. π

11. If the length of the wire of a sonometer is halved, the value of the resonant frequency will be

a. double b. half c. four times d. eight times

12. A thin lens of focal length f has aperture d. It forms an image of intensity I, Inner part of lens up to diameter d/3 is painted black. The intensity of image will be

a. $\frac{I}{3}$ b. $\frac{I}{9}$ c. $\frac{8I}{9}$ d. I

13. A lens is placed between a source of light and a wall. If forms images of area A_1 and A_2 on the wall for its two different positions. The area of the source of light is

a. $\sqrt{A_1A_2}$ b. $\frac{A_1 + A_2}{2}$ c. $\left(\frac{1}{A_1} + \frac{1}{A_2}\right)^{-1}$ d. $\left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2}\right)^2$

14. Young's experiment is performed with light of wavelength 6000Å wherein 16 fringes occupy a certain region of the screen. If 24 fringe occupy the same region with another light of wavelength λ, then λ is

a. 6000Å b. 4500Å c. 5000Å d. 4000Å

15. The electric strength of air is 2×10^6 N/c. The largest charge that a metallic sphere of 3mm radius can hold is

a. 2nc b. 3nc c. 1.5nc d. 6nc

16. If the electric flux entering and leaving an enclosed surface respectively are ϕ_1 and ϕ_2, the electric charge inside the surface will be

a. $\frac{\phi_1 + \phi_2}{\varepsilon_0}$ b. $\frac{\phi_2 - \phi_1}{\varepsilon_0}$ c. $(\phi_1 + \phi_2)\varepsilon_0$ d. $(\phi_2 - \phi_1)\varepsilon_0$

17. An electron is moving in a circle of radius 5.1×10^{-11}m at a frequency of 6.8×10^{15}rev/sec. The equivalent current is approximately

a. 1.1×10^{-3}A b. 2.2×10^{-3}A c. 3.3×10^{-3}A d. 4.4×10^{-3}A

18. A helium nucleons makes full rotation in a circle of radius 0.8m in 2sec. The value of magnetic field B at the centre of the circle will be

a. $10^{-19}\mu_0$ b. $2 \times 10^{-19}\mu_0$ c. $\frac{10^{-19}}{\mu_0}$ d. $2 \times 10^{-20}\mu_0$

19. The universal property among all substance is

a. diamagnetism b. para magnetism
c. ferromagnetism d. non-magnetism

20. The deflection in a moving coil galvanometer is reduced to half when it is shunted with a 40Ω coil. The resistance of the galvanometer is

a. 80Ω .b. 40Ω c. 20Ω d. 15Ω

21. In an A.C. circuit V and I are given by V = 100sin(1000t) volts and I = 1000sin(1000t + π/3) mA. The power dissipated in the circuit is

a. 10^4w b. 25w c. 10w d. 250w

22. The impedance is $\sqrt{3}$ times of resistance. The phase difference is

a. 45^0 b. 30^0 c. 60^0 d. 75^0

23. Which of the following have highest specific charge?
    a. position b. proton c. He c. none of above
24. The de-Broglie wavelength of the electron in the ground state of the hydrogen atom is (radius of 1st orbit = 0.53Å)
    a. 1.67Å b. 3.35Å c. 1.06Å d. 1.53Å
25. When a hydrogen atom emits a photon of energy 12.1eV its orbital angular momentum changes by
    a. 1.05×10^{-34} J-s b. 2.11×10^{-34} J-s
    c. 3.16×10^{-34} J-s d. 4.22×10^{-34} J-s
26. The nucleus of element X (A = 220) undergoes an α-decay. If Q value of the reaction is 5.5MeV, then approximate KE of α-particle is
    a. 5.5MeV b. 5.4MeV c. 5.3MeV d. 5MeV
27. In a semiconductor material the mobilities of electron and hole are μ_e and μ_n respectively. Which of the following is true?
    a. $\mu_e > \mu_n$ b. $\mu_e < \mu_n$ c. $\mu_e = \mu_n$ d. $\mu_e < 0, \mu_n > 0$
28. A fireman wants to slide down a rope. The breaking load for the rope is $\left(\frac{3}{4}\right)^{th}$ of the weight of the man. With what acceleration should the fireman slide down.
    a. $\frac{g}{4}$ b. $\frac{g}{2}$ c. $\frac{3g}{4}$ d. zero
29. Two cars are moving in the same direction with the same speed of 30 km/hr. They are separated by 5 km. What is the speed of the car moving in the opposite direction if it meets the two cars at an interval of 4 minute?
    a. 15 km/hr b. 30 km/hr c. 45 km/hr d. 60 km/hr
30. By what percent the energy of a satellite has to be increased to shift it from an orbit of radius r to $\frac{3r}{2}$?
    a. 15% b. 20.3% c. 66.7% d. 33.33%

Subject: Chemistry

Time: 1 hr.

1. The specific heat of metal is 0.11 and its equivalent weight is 18.61. Its exact atomic weight is.
    a. 58.2 b. 29.1 c.55.83 d.27.91
2. Which of these is correct set of quantum numbers for a 4d electron?
    a. 4,3,2,+1/2 b.4,2,2,-1/2 c.4,2,1,0 d.4,0,1,-1/2
3. In the following reaction, the missing particle is. ${}^{6}_{3}Li + ? \rightarrow {}^{4}_{2}He + {}^{3}_{1}H$
    a. electron b. neutron c. proton d. deuteron
4. Which of these have highest second ionization potential?
    a. C b. N c. O d.F

5. The hybridization on Xe atom in XeF_2 is.
   a.sp^3 b.sp^3d c.sp^3d^2 d. dsp^2
6. Which of these is most acidic?
   a.BF_3 b.BCl_3 c.BBr_3 d.BI_3
7. The number of particles in bcc unit cell is.
   a.1 b.2 c.3 d.4
8. The normality of 0.3M phosphorous acid$(H_3PO)_3$. is.
   a.0.3 b.0.6 c.0.9 d.o.1
9. The partial pressure of O_2 in the reaction is. $2Ag_2O(s) \longleftrightarrow 4Ag(s) + O_2(g)$
   a. K_p b. $\sqrt{K_p}$ c.$(K_p)^{1/3}$ d.$2K_p$
10. Which of this solution has highest pH value?
    a.CH_3COONH_4 b.Na_2CO_3 c.NH_4Cl d.$NaNO_3$
11. The difference between heats of reaction at constant pressure and constant volume for the reaction
    $2C_6H_6(l) + 15O_2(g) \longrightarrow 12CO_2(g) + 6H_2O(l)$ at 25^oC in Kj is.
    a.-7.43 b.+3.72 c.-3.72 d.+7.43
12. The rate law expression of a reaction is Rate=$K[A]^n[B]^m$. on doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as.
    a. $\frac{1}{2^{m+n}}$ b. m+n c. n-m d.2^{n-m}
13. The maximum precipitating power for arsenic sulphide sol is of.
    a. K_2SO_4 b.$CaCl_2$ c.Na_3PO_4 d.$AlCl_3$
14. What will be the emf of the following cell? Pt $IH_2(P_1)IH^+(aq)IH_2(P_2)IPt$.
    a. $\frac{RT}{F}\log\frac{P_1}{P_2}$ b. $\frac{RT}{2F}\log\frac{P_1}{P_2}$ c. $\frac{RT}{F}\log\frac{P_2}{P_1}$ d. $\frac{RT}{2F}\log\frac{P_2}{P_1}$
15. The method used to purify Ni using CO is called
    a. Baeyer's process b. Parke's process
    c. Van-Arkel's process d. Mond's process
16. A compound when mixed with water gives hissing sound and becomes very hot. The compound is.
    a.$CaCl_2$ b. $CaCO_3$ c. CaO d.$Ca(OH)_2$
17. $FeSO_4$ forms brown ring with.
    a.NO_3 b.NO_2 c. NO d.N_2O_3
18. Iodine oxidizes $S_2O_3^{--}$ ion to
    a.SO_3^{2-} b.SO_4^{2-} c.S^{2-} d.$S_4O_6^{2-}$
19. Which of these is most volatile compound?
    a. HF b. HC*l* c. HBr d. HI

20. Iron becomes passive on reaction with

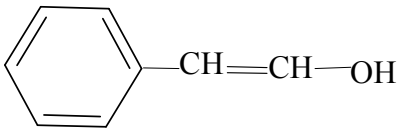
a. conc. HC*l* b. conc. H_2SO_4 c. conc.H_3PO_4 d. conc. HNO_3

21. The type of hybridization of nickel in the complex $Ni(CO)_4$ is.

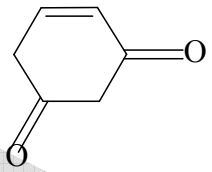
a.sp^3 b.dsp^2 c. sp^3d d.dsp^3

22. Tautomerism is exhibited by.

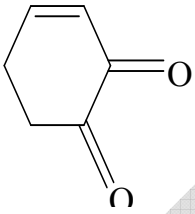
a.



b.



c.



d. all of these

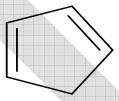
23. The name of the organic compound given below is.

a. 3-methyl-4-ethyloctane b.2,3-diethylheptane

c.5-ethyl-6-methyloctane d.4-ethyl-3-methyloctane

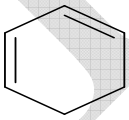
24. The ozonolysis product of an olefin are OHC-CHO and OHC-CH_2CH_2-CHO, the olefin is.

a)



b)

c)



d)

25. The reaction, $2C_6H_5I + Cu \longrightarrow C_6H_5\text{-}C_6H_5$ is called

a. Wurtz's reaction b. Fittig reaction c. Wurtz-fittig reaction d. Ullmann reaction

26. Which of these is used to detect alcohol in the Breathalyzer tube used to find alcohol intoxication of drivers?

a.$Cr_2O_7^{--}$ b.Cr_2O_3 c.MnO_2 c.$KMnO_4$

27. In the reaction: $C_6H_5CHO + C_6H_5NH_2 \longrightarrow C_6H_5NH{=}CHC_6H_5 + H_2O$. The product formed is.

a. Aldol b. Schiff's reagentc. Schiff's base d. Benedict's reagent

28. Insulin, a hormone is

a. fat b. steroid c. protein d. carbohydrate

29. Primary, secondary and tertiary amines are separated using
    a. Benzene sulphonyl chloride b. diethyl succinate
    c.Sulphonyl chloride d. acetyl chloride
30. $CH_3CH_2CONH_2 \xrightarrow{X} CH_3CH_2CH_2NH_2$, x is
    a. Pt/H_2 b.Ni/H_2 c.$LiAlH_4$ d. Zn

Subject: English

Time: 1 hr.

Choose the right choice(Q. no 1-9)

1. She has knack _____ solving difficult problems
   a. at b. for c. with d. of
2. I am fed up _____the same food daily.
   a. off b. with c. in d. over
3. His company is greatly sought _____
   a. from b. of c. after d. for
4. He is born _____rich parents
   a. for b. of c. off d. by
5. He is bent _____passing
   a. to b. on c. at d. over
6. The expression " Get on my nerves' means
   a. disappoint b. get worried c. disturb d. leave alone
7. " Red handed" means
   a. quickly b. caught in the act
   c. caught with dirty hands d. caught with his hands tied
8. " Wild goose chase" means
   a. wise decision b. useless search c. timely action d. delayed action
9. " Flogging a dead horse" means
   a. a settled matter b. over spending c. going too far d. wasting one's energy

Choose the right alternative (Q. no. 10-12)

10. Brush: Paint
    a. trowel: cement b. polish: floor c. rake: lawn d. match: fire
11. Sheep: wool
    a. otter: fur b. fodder: animal c. flax: cotton d. Stupid: good
12. Bushels: Potatoes
    a. ounce: coal b. wood: cord c. point: diamonds d. bricks: mortar

Choose the appropriates synonym (Q. no. 13-17)

13. Incumbent
    a. odd b. obligatory c. blatant d. subdue
14. Imbibe
    a. engross b. baffle c. serene d. laxity
15. Grandeur
    a. greatness b. silent c. gross d. liability
16. Gullible
    a. credulous b. unlucky c. entangle d. subdue
17. Explicit
    a. expressed b. loss c. prudence d. spotted

Choose the right antonym (Q. no 18 -20)

18. Obnoxious
    a. harmful b. lovable c. legend d. dissolve
19. Malice
    a. spite b. detest c. praise d. abate
20. Juvenile
    a. young b. energetic c. bankrupt d. indifferent

Choose the right choice (Q.no.21-26)

21. Money paid to divorced wife
    a. remuneration b. alimony c. sinecure d. ex-gratis
22. A speech made without preparation
    a. eloquent b. extempore c. verbose d. maiden
23. A religious discourse
    a. sermon b. stanza c. preach d. sanctorum
24. One who has a habit of stealing
    a. kleptomaniac b. poacher c. thief d. bandit
25. A person who hates women
    a. hater b. rapist c. misogynist d. misogamist
26. A large departure of people
    a. exodus b. immigration c. emigration d. epidemic

Choose the part of the sentence that is grammatically incorrect (Q.no 27- 30)

27. <u>This property</u> was <u>to be</u> divided <u>between all</u> <u>the brothers</u>
    a. b. c. d.
28. <u>The girl</u> <u>resembles</u> <u>to her</u> parents. <u>no error.</u>
    a. b. c. d.
29. <u>She missed</u> <u>to catch</u> her <u>flight to</u> London <u>on time</u>
    a. b. c d.
30. <u>It is</u> <u>them who</u> <u>are responsible</u> for <u>the carelessness</u>.
    a. d. c. d.

Subject: Mathematics

Time: 1 hr.

1. The value of i^i is

   a. $e^{\pi/2}$ b. $e^{-\pi/2}$ c. 2i d. Not define

2. The number of 3 × 3 matrices A whose entries are either 0 or 1 and for which the system A $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ has exactly two distinct solutions is

   a. 0 b. $2^9 - 1$ c. 168 d. 2

3. The greatest number among $\sqrt[3]{9}, \sqrt[4]{11}, \sqrt[6]{17}$ is

   a. $\sqrt[3]{9}$ b. $\sqrt[4]{11}$ c. $\sqrt[6]{17}$ d. Can't be determine

4. If second term in the expansion $\left[\sqrt[13]{a} + \frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$ then the value of $\frac{nC_3}{nC_2}$ is

   a. 4 b. 3 c. 12 d. 6

5. If the third term of a G.P. is 4, then the product of its 1st five terms is

   a. 4^3 b. 4^4 c. 4^5 d. $4^{1/5}$

6. If f(x) = cos(logx), then f(x). $f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ has the value

   a. –1 b. $\frac{1}{2}$ c. –2 d. 0

7. The domain of definition of the function $f(x) = \sqrt{1 + \log_e (1 - x)}$ is

   a. $-\infty < x \le 0$ b. $-\infty < x \le \frac{e-1}{e}$

   c. $-\infty < x \le 1$ d. $x \ge 1 - e$

8. If f(x) = (x – 2) (x – 4) (x – 6) …… (x – 2n), then f ′(2) is

   a. $(-1)^n 2^{n-1} (n-1)!$ b. $(-2)^{n-1} (n-1)!$

   c. $(-2)^n n!$ d $2^{n-1}(n-1)!$

9. The smallest circle with centre on y-axis and passing through the point (7, 3) has radius

   a. $\sqrt{78}$ b. 7 c. 3 d. 4

10. $\int e^x \left(\frac{x-1}{x^2}\right)$

    a. $\frac{e^x}{x^2} + c$ b. $-\frac{e^x}{x^2} + c$ c. $\frac{e^x}{x} + c$ d. $\frac{e^{-x}}{x} + c$

11. If $y = \cos\frac{x}{2} . \cos\frac{x}{4} . \cos\frac{x}{8} \ldots\ldots \cos\frac{x}{2^n}$, then the value of y at $n \to \infty$ is

a. $\frac{\sin x}{x}$ b. $\frac{x}{\sin x}$ c. $e^x.\sin x$ d. $x \cos\frac{x}{2}$

12. If f(x) is a function satisfying $f\left(\frac{1}{x}\right) + x^2 f(x) = 0$ for all non-zero x, then $\int_{\sin\theta}^{\operatorname{cosec}\theta} f(x)\,dx$ is equal to

a. 0 b. 1 c. 2 d. 3

13. The area bounded by $y = |\sin x|$, x-axis and the lines $|x| = \pi$ is

a. 2 sq. unit b. 3 sq. unit c. 4 sq. unit d. 5 sq. unit

14. In order to eliminate the first degree terms from the equation $2x^2 + 4xy + 5y^2 - 4x - 22y + 7 = 0$, the point to which origin is to be shifted is

a. (1, –3) b. (2, 3) c. (–2, 3) d. (1, 3)

15. The line $ax + by + c = 0$ is normal to the circle $x^2 + y^2 + 2gx + 2fy + d = 0$ if

a. $ag + bf + c = 0$ b. $ag + bf - c = 0$

c. $ag - bf + c = 0$ d. $ag - bf - c = 0$

16. The equation $\frac{x^2}{2-\lambda} - \frac{y^2}{\lambda-5} - 1 = 0$ represents an ellipse, if

a. $\lambda > 5$ b. $\lambda < 2$ c. $2 < \lambda < 5$ d. $2 > \lambda > 5$

17. The value of $\left[\vec{i}\ \vec{j}\ \vec{k}\right] + \left[\vec{j}\ \vec{k}\ \vec{i}\right] + \left[\vec{k}\ \vec{j}\ \vec{i}\right]$ is

a. 1 b. 2 c. 3 d. 0

18. From the top of a tower, the angle of depression of a point on the ground is 60^0. If the distance of this point from the tower is $\frac{1}{\sqrt{3}+1}$, then the height of the tower is

a. $\frac{4\sqrt{3}}{2}$ m b. $\frac{\sqrt{3}+3}{2}$ m c. $\frac{3-\sqrt{3}}{2}$ m d. $\frac{\sqrt{3}}{2}$ m

19. If angles A, B, C are in A.P., then $\frac{a+c}{b}$ is equal to

a. $2 \sin\frac{A-C}{2}$ b. $2 \cos\frac{A-C}{2}$ c. $\cos\frac{A-C}{2}$ d. $\sin\frac{A}{2}$

20. The probability that a boy will get a scholarship is 0.75 and that a girl will get is 0.72. What is the probability that at least one of them will get the scholarship?

a. 0.83 b. 0.93 c. 0.95 d. 0.98

21. For a sphere of radius 9 cm, the numerical value of the volume expressed as a percentage of numerical value of the surface is:

a. 300% b. 200% c. 150% d. 3%

22. a, b, c, d, e are five consecutive numbers, their average is

a. a + 2 b. $\frac{abcde}{2}$ b. b d. $\frac{c+d}{2}$

23. By selling oranges 12 for Rs. 10, vendor loses 20%. In order to gain 20%, the number of oranges he should sell for Rs.10 is

a. 8 b. 7 c. 6 d. 5

24. A regular polygon of nine sides, each of length 2, is inscribed in a circle. The radius of circle is

a. $\cos\frac{\pi}{9}$ b. $\frac{1}{2}\cot\frac{\pi}{9}$ c. $\frac{1}{2}\operatorname{cosec}\frac{\pi}{9}$ d. $\operatorname{cosec}\frac{\pi}{9}$

25. 6 men earn as much as 7 women, 2 women as much as 3 boys and 3 boys as much as 4 girls. If a girl earns Rs. 18, what does a man earn?

a. 40 rupees b. 41 rupees c. 42 rupees d. 44 rupees

26. For a 40 hours a week Sachin is paid Rs. 40 as basic pay. Overtime is paid at 25% above the basic rate. In a certain week Sachin worked overtime and his total wages was Rs. 50. Sachin therefore worked for a total of:

a. 42 hours b. 52 hours c. 48 hours d. 58 hours

27. If S = {1, 2, 3, 4}. The total number of unordered pairs of disjoint subsets of S is equal to

a. 25 b. 34 c. 42 d. 41

28. A and B enter into a partnership for a year. A contributes Rs. 1500 and B Rs. 2000. After 4 months, they admit C who contributes Rs. 2250. If B withdraws his contribution after 9 months, the profit at the end of the year should be shared in the ratio:

a. 1 : 1 : 1 b. 2 : 1 : 1 c. 1 : 2 : 1 d. 1 : 2 : 3

29. If a polygon has same number of sides as its diagonals, then the number of sides is

a. 5 b. 6 c. 7 d. 8

30. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z + \cos^{-1}t = 4\pi$, then the value of $x^2 + y^2 + z^2 + t^2$ is

a. xy + yz + zt b. 4 c. 1 – 2xyzt d. 6

…The End…

Answers: Set II

Subject: Physics

1.a	2.c	3.b	4.a	5.d	6.c	7.c	8.a	9.b	10. c
11. a	12. c	13. a	14. d	15. a	16. c	17. a	18. a	19. a	20. b
21. b	22. a	23. a	24. b	25. b	26. b	27. a	28. a	29. c	30. d

Subject: Chemistry

1.c	2.b	3.b	4.c	5.b	6.d	7.b	8.b	9.a	10. b
11. a	12. d	13. d	14. b	15. d	16. c	17. c	18. d	19. b	20. d
21. a	22. d	23. d	24. c	25. d	26. a	27. c	28. c	29. a	30. c

Subject: English

1.b	2.b	3.c	4.b	5.b	6.c	7.b	8.b	9.d	10. a
11. a	12. c	13. b	14. a	15. a	16. a	17. a	18. b	19. c	20. c
21. b	22. b	23. a	24. a	25. c	26. a	27. c	28. c	29. b	30. b

Subject: Mathematics

1.b	2.a	3.a	4.a	5.c	6.d	7.b	8.b	9.b	10. c
11. a	12. a	13. c	14. c	15. b	16. b	17. a	18. c	19. b	20. b
21. a	22. a	23. a	24. d	25. c	26. c	27. d	28. a	29. a	30. c

PEA

EDUCATION WING
EMBASSY OF INDIA
KATHMANDU, NEPAL
BE. Model Entrance Examination

Attempt all Questions. Each question carries one mark.

Choose the correct answer and blacken the appropriate bubble using HB pencil on answer sheet.

SET- III

Subject: Physics

Time: 1 hr.

1. A bar magnet having magnetic dipole moment M is bent in the form of circular arc such that arc subtends angle θ at centre. Find new dipole moment.

   a. $\frac{M \sin\frac{\theta}{2}}{\theta}$ b. $\frac{2M \sin\frac{\theta}{2}}{\theta}$ c. $\frac{2M \sin\theta}{\theta}$ d. $\frac{2M \cos\frac{\theta}{2}}{\theta}$

2. If real dip at a place is 30^0 then find apparent dip at that place (possible answer).

   a. 30^0 b. 25^0 c. 20^0 d. 45^0

3. If work done in rotating a dipole through 60^0 is W then find work done in rotating through 180^0

   a. W b. 2W c. 3W d. 4W

4. Two metallic spheres have charges $+q_1$, and $+q_2$ respectively. They are r distance apart in free space. If F be coulomb's force between them then

   a. $F = K.\frac{q_1q_2}{r^2}$ b. $F > K.\frac{q_1q_2}{r^2}$ c. $F < K.\frac{q_1q_2}{r^2}$ d. none of the above

5. Electric potential due to a positive charge

   a. is always positive b. is always negative
   c. is always zero d. May be + ve, –ve or zero.

6. A proton is thrown towards the nucleus of an atom with speed *v* and closest distance of approach is r. If the same proton is thrown towards the same nucleus with speed 2*v* then closest distance of approach will be

   a. 2r b. $\frac{r}{2}$ c. $\frac{r}{4}$ d. r

7. Capacitor is

   a. Current storing device b. Energy storing device
   c. Energy converter d. All of the above

8. A copper wire of resistance 0.5Ω/m, now it is bent in the form of circle of radius 1m, same wire is connected along its diameter then equivalent resistance between opposite ends of diameter is

   a. $\frac{\pi+4}{4}$ b. $\frac{\pi}{\pi+4}$ c. $\frac{\pi}{4}$ d. $\frac{\pi+4}{\pi}$

9. The electric bulbs 25W, 250V and 100W, 250V are connected in series with 500V supply. Which bulb will fuse

   a. 25 W only b. 100 W only c. both d. none of them

10. Which is immaterial for fuse wire

a. length b. resistivity c. density d. none of the above

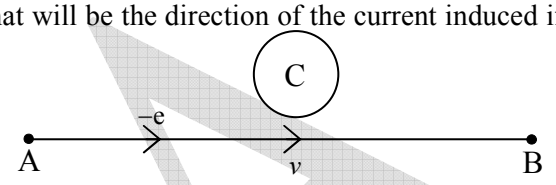
11. The ratio of magnetic field at the centre of circular coil and the centre of square for same current having same length is

a. $\pi^2 : 8\sqrt{2}$ b. $8\sqrt{2} : \pi^2$ c. $\pi^2 : 2\sqrt{2}$ d. $8\sqrt{2} : \pi^2$

12. Rms value of an alternating voltage given by $e = e_1 \cos\omega t + e_2 \sin\omega t$, is

a. $\frac{e_1 + e_2}{2}$ b. $\sqrt{e_1 e_2}$ c. $\sqrt{\frac{e_1^2 + e_2^2}{2}}$ d. $\sqrt{e_1^2 + e_2^2}$

13. An electron moves only the line AB which lies in the same plane as a circular loop of conducting wire as shown in figure. What will be the direction of the current induced if any in the loop ?



a. no current will be induced

b. the current will be clockwise

c. the current will be anticlockwise

d. the current will change direction as the electron passes by

14. When a light wave suffers reflection at the interface from air to glass, the change in phase of the reflected wave is equal to

a. zero b. $\frac{\pi}{2}$ c. π d. 2π

15. Two Polaroid's as oriented with their planes perpendicular to incident light and transmission axis making an angle of 30^0 with each other. What fraction of incident unpolarised light is transmitted?

a. 10 % b. 37.5 % c. 62.5 % d. none of these

16. A proton and α–particle projected with equal linear momenta describe circular path of radii r_1 & r_2 respectively in uniform magnetic field. The ratio of r_1 & r_2 is

a. 1 b. 2 c. $1 : 2\sqrt{2}$ d. 1 : 2

17. Photo electric effect is due to

a. Free electrons b. Bound electrons

c. Protons d. Neutrons

18. If M is the mass of a nucleus and A its atomic mass, then the packing fraction is

a. $\frac{M - A}{M + A}$ b. $\frac{M - A}{M}$ c. $\frac{M - A}{A}$ d. $\frac{M + A}{M - A}$

19. When the temperature of intrinsic semiconductor is increased

a. Resistance of semiconductor increases

b. The heat energy decreases the atomic radius

c. The holes are created in conduction bond

d. The energy of atoms is increased

20. A particle & anti particle fuse together and energy formed is E. Find mass of either particle

a. $\frac{E}{C^2}$ b. $\frac{E}{2C^2}$ c. $\frac{2E}{C^2}$ d. $\frac{E}{2C}$

21. A nail is located at a certain distance vertically below the point of suspension of a simple pendulum. The pendulum bob is released from the position where the string makes angle of 60^0 from the vertical. Calculate the distance of the nail from the point of suspension such that the bob will just perform revolution with the nail as centre. Assume length of pendulum 1 m.

a. 0.8 m b. 0.5 m c. 0.2 m c. 0.6 m

22. A number of particles each having mass 'm' is placed along +ve x-axis at distances r, 2r 4r, 8r, …… from origin. Find net gravitational potential at origin.

a. $-\frac{Gm}{r}$ c. $-\frac{4}{3}.\frac{Gm}{r}$ c. $-\frac{2Gm}{r}$ d. $-\frac{Gm}{2r}$

23. A particle in SHM has frequency 'f '. Find frequency of K.E.

a. f b. 2f c. $\frac{f}{2}$ d. zero

24. A cubical block is floating in water with 'h' depth immersed. If the system is moved vertically upward with acceleration then immersed depth is H.

a. h = H b. h < H c. h > H d. can't be predicted

25. A body of mass 2kg is acted by a force such that $x = \frac{t^3}{3}$. Where x is in meter & t in seconds. Find work done by force in first 2 seconds.

a. 2 J b. 4 J c. 8 J d. 16 J

26. Breaking stress for a wire of radius r is s. Find breaking stress another wire of same substance but radius 2r.

a. S b. 4S c. $\frac{S}{4}$ d. 2S

27. A heat engine works on a Carnot cycle with a heat sink at a temperature of 2700K. The efficiency of the engine is 10%. Determine the temperature of heat source.

a. 300 K b. 3000 K c. 1500 K d. 2700 K

28. When a floating ice block melts in water then level of water

a. increase b. decrease

c. remains same d. first increases then decreases

29. The characteristic by which two sound waves of same frequency and loudness be distinguished is

a. intensity b. pitch c. quality d. amplitude

30. When two tuning forks A & B are sounded together x beats/sec are heard. Frequency of A is n. Now when one prong of B is loaded with a little wax the number of beats per second decreases. The frequency of fork B is

a. n + x b. n – x c. $n - x^2$ d. n – 2x

Subject: Chemistry

Time: 1 hr.

1. One mole of CO_2 contains
   a. 6.02×10^{23} atoms of C b. 6.02×10^{23} atoms of O
   c. 18.1×10^{23} molecules of CO_2 d. 1 gram atoms of oxygen
2. Transition of an electron from n = 3 level to n = 1 level results in
   a. x – ray spectrum b. Emission spectrum
   c. Band spectrum d. IR spectrum
3. ${}_7N^{13}$ changes to ${}_6C^{13}$ by emission of
   a. electron b. neutron c. positron d. proton
4. The correct order of electron affinity value is
   a. F < C*l* < Br < I b. I < Br < F < C*l*
   c. I < Br < C*l* < F d. Br < I < C*l* < F
5. Which of these species is paramagnetic?
   a. N_2 b. F_2 c. O_2^- d. O_2^{--}
6. The correct set of oxidation numbers of nitrogen in ammonium nitrate is
   a. –3, +3 b. –1, +1 c. +1, –1 d. –3, +5
7. The volume of 0.05M $KMnO_4$ solution required to oxidize 2g of $FeSO_4$ in acidic medium is (Mol. wt. of $FeSO_4$ = 152)
   a. 52.63 m*l* b. 5.263 m*l* c. 64.82 m*l* d. 6.48 m*l*
8. An ionic compound has a unit cell consisting of ions A at the corners of a cube and ions B at the faces of the cube. The formula of the compound would be.
   a. AB b. A_2B c. AB_3 d. A_3B
9. The concentration in grams per litre of a solution of cane sugar (mol. wt = 342) which is isotonic with a solution containing 6 grams of urea (mol. wt = 60) per litre is.
   a. 3.42 b. 34.2 c. 5.7 d. 19
10. One mole of SO_3 was placed in a litre reaction vessel at a certain temperature. The following equilibrium was established $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$. At equilibrium, 0.6 moles of SO_2 were formed. The equilibrium constant of the reaction will be.
    a. 0.365 b. 0.455 c. 0.545 d. 0.675
11. If an endothermic reaction is non-spontaneous at freezing point of water and becomes feasible at its boiling point, then
    a. ΔH = –ve, ΔS = +ve b. ΔH and ΔS are +ve
    c. ΔH and ΔS are –ve d. ΔH = +ve, ΔS = –ve
12. For a zero order reaction, A → products, $t_{1/2}$ is.
    a. $\frac{[A]_o}{2K}$ b. $\frac{ln2}{K}$ c. $\frac{1}{K[A]_o}$ d. $\frac{ln2}{[A]_oK}$
13. A colloid formed by dispersion of liquid in solid is
    a. Gel b. Sol c. emulsion d. foam

14. When current is passed through Al_2O_3, 13.5 g of Al are deposited, the number of Faraday required is.
    a. 0.5 b. 1 c. 1.5 d. 2
15. The metal extracted by leaching of cyanide is.
    a. Mg b. Ag c. Cu d. Na
16. Which of the doesn't give flame test?
    a. Na b. Li c. Mg d. Ca
17. Aluminium is extracted by electrolysis of a mixture of Al_2O_3 and
    a. Alumina b. Cryolite c. Bauxite d. Felspar
18. Which variety of phosphorous burns spontaneously in air?
    a. White b. Red c. Black d. Scarlet
19. Which of the following is oxidized by SO_2?
    a. Mg b. $K_2Cr_2O_7$ c. $KMnO_4$ d. all
20. I_2 dissolves readily in KI solution giving.
    a. I^- b. KI_2 c. KI_3^- d. KI_3
21. The type of isomerism present in nitropentammine chromium (III) chloride is.
    a. optical b. linkage c. ionization d. polymerisation
22. The Beilstein test for organic compounds is used to detect
    a. carbon b. halogen c. nitrogen d. sulphur
23. Which of the following has heighest nucleophilicity?
    a. F^- b. OH^- c. CH_3^- d. NH_2^-
24. The knocking property of Diesel is expressed in terms of.
    a. octane number b. cetane number
    c. oxytane number d. ocetane number
25. The reaction $C_6H_5N_2Cl \xrightarrow{Cu+HBr} C_6H_5Br$ is.
    a. Sand Meyer's reaction b. Gattermann's reaction
    c. Gattermann–koch's reaction d. Balz Schiemann reaction
26. Methyl ketones are usually characterized through
    a. Tollen's reagent b. Iodoform test c. Schiff's test d. Benedict's reagent
27. 1^0, 2^0 and 3^0 amines may be separated using.
    a. Benzene sulphonyl chloride b. Diethyl succinate
    c. Sulphonyl chloride d. Acetyl chloride
28. Ascorbic acid is
    a. vitamin b. enzyme c. protein d. none
29. Substances which bring down body temperature are known as
    a. Antipyretics b. Analgesics c. Antibiotics d. None of these
30. Formic acid and acetic acid can be distinguished using.
    a. Litmus b. $NaHCO_3$ c. Caustic soda d. Ammonical $AgNO_3$

Subject: English

Time: 1 hr

Directions: **Go through the passage and answer the questions that follow: (1-5)**

There is no general agreement about how the planets were formed. The most widely accepted theory is that about 5, 000 million years ago swirling clouds of matter began to condense. Through the action of a centrifugal force, the heavier molecules were concentrated near the centre of the eddies, and the lighter gaseous material was thrown out the periphery. Such is the theory. What is known is that nine satellites began orbiting round the sun. These are the planets.

The planet on which man lives is the third closest to the sun, with the third shortest orbit. It also has something none of the others has – an atmosphere that can support life in all the manifold forms that exist on our planet. There may be satellites circling others stars in other parts of the universe, which have the right ingredients for some sort of life to evolve, but the earth is the only one in the solar system.

1. The theory of the formation of the planets
   a. is a debatable one
   b. is generally agreed upon by everyone
   c. covers a very wide area
   d. is fairly well known
2. According to the passage, the planets are
   a. nothing but condensed clouds
   b. a collection of gaseous material
   c. a collection of condensed swirling material
   d. a collection of centrifugal forces
3. One essential difference between the earth and the other planets is that
   a. the atmosphere of the earth makes possible the presence of life on it
   b. the earth draws the heavier molecules into its centre through the action of centrifugal force
   c. only the earth is on the periphery of its solar system
   d. the earth has the capacity to come into closer contact with the sun
4. The writer claims that the life –supporting atmosphere
   a. is there on other planets in the solar system
   b. maybe there are other satellites in the universe
   c. may evolve on other satellites encircling other stars in the universe
   d. cannot evolve anywhere outside the earth
5. A planet is a heavenly body which moves round
   a. the sun
   b. a star
   c. a satellite of the solar system
   d. the universe

Choose the right synonym (6 -10)

6. Callous
   a. large
   b. hardened
   c. thick
   d. dangerous
7. Incite
   a. discourage
   b. encourage
   c. internal
   d. crime

8. Grudge
   a. hard b. hot c. rare d. envy
9. Accomplice
   a. friend b. assistant c. criminal d. disloyal
10. Candid
   a. sweet b. pleasing c. frank d. vigor

Rearrange the jumbled parts to make proper sense. 1 and 6 have been done for you and are in proper order, then choose the right alternative

1. Thirty-seven years later
P. to reiterate that the highest Q. to farming, to food and agriculture
R. the time has come for us S. national priority must be given
6. as the corner stone of the economy

11. a. RSPQ b. SQPR
   c. RPSQ d. QPSR

1. Second, we shall include in the
P. Labor input of the farmers Q. the managerial and
R. at a higher wage, reflecting S. cost of production the
6. entrepreneurial role of the farmer

12. a. QRPS b. RQPS
   c. SPRQ d. QPRS

Choose the right antonym (13 -15)

13. Deleterious
   a. impulsive b. support c. pathetic d. inclusive
14. Eschew
   a. traduce b. invite c. use d. emanate
15. Savory
   a. sad b. hidden c. lost d. unpalatable

Choose the right choice (16-25)

16. I was _____by her spontaneous replies during the beauty pageant.
   a. carried off b. carried away c. carried out d. carried over
17. 'To have an axe to grind' means
   a. to pinpoint failures b. to take revenge
   c. to make demands d. selfish interest to serve
18. 'A pipe dream' means
   a. daydream b. hope or plan which will not materialize
   c. an unending dream d. a nightmare
19. The lease of the shop <u>ran out</u>.
   a. expired b. appeared c. supported d. postponed

20. She can no longer _____ her husband.
    a. bear upon b. bear with c. bear by d. bear out
21. She groped _____the dark
    a. at b. into c. out d. in
22. Neither of the two examples _____ appropriate
    a. seem b. have seemed c. seems d. none
23. He came to Kathmandu after he _____ to be the manager of the company
    a. ceased b. has ceased c. was ceased d. had ceased
24. Nepal economy grew _____6% in the fiscal year
    a. with b. along c. in d. by
25. Please do not get on my nerves, I am in a crisis
    a. disappoint b. to add more worry
    c. disturb d. leave alone

Choose the right analogies (26-30)

26. Amplify: Augment
    a. redouble : efforts b. exacerbate: sale
    c. maximize: profit d. enhance: power
27. Horse: Stable
    a. sty: pig b. coop: hen c. fish : pond d. goats: pen
28. Ink: Pen
    a. water: river b. syrup: juice c. blood: body d. satellite: constellation
29. Ship: dockyard, Money: _____
    a. wealth man b. poor man c. cashier d. mint
30. Pitch: Cricket. Rink: _____
    a. wrestling b. badminton c. boxing d. skating

Subject: Mathematics

Time: 1 hr.

1. Which one of the following is $(A - B) \cup (B - A)$?
   a. $(A\cup B) \cup (A - B)$ b. $(A\cup B) \cup (A\cap B)$
   c. $(A\cup B) - (A\cap B)$ d. $(A - B) \cap (B - A)$
2. Which of the following is odd function?
   a. $f(x) = \frac{a^x + a^{-x}}{a^x - a^{-x}}$ b. $f(x) = \cos x$ c. $f(x) = \frac{a^x + 1}{a^x - 1}$ d. $f(x) = \sin^2 x$
3. value of $\lim_{x\to 0} \frac{e^{\tan x} - e^x}{\tan x - x}$ is:
   a. 0 b. 1 c. e d. e –1
4. for the curve $2^x+2^y = 2^{x+y}$, the slope of tangent at (1,1) is
   a. -1 b. 0 c. 1 d. tangent is parallel to y-axis

5. the function f(x) = 1+ [cosx]x in $0< x < \frac{\pi}{2}$:

a. has a minimum value 0 b. has maximum value 2

c. is continuous but not differential d. is continuous and differential

6. $\int e^{\frac{x}{2}} \sin\left(\frac{x}{2}+\frac{\pi}{4}\right)dx$ is

a. $\sqrt{2}e^{\frac{x}{2}} \sin\frac{x}{2}+c$ b. $\sqrt{2}e^{\frac{x}{2}} \cos\frac{x}{2}+c$

c. $\frac{1}{\sqrt{2}}e^{\frac{x}{2}} \sin\frac{x}{2}+c$ d. $\frac{1}{\sqrt{2}}e^{\frac{x}{2}} \cos\frac{x}{2}+c$

7. $\int_0^{2\pi} \sqrt{1+\sin\frac{x}{2}}\, dx =$

a. 0 b. 2 c. 4 d. 8

8. The area bounded by y = cosx, $x = \frac{-\pi}{2}$ and $x = 2\pi$ and the axis of x in square units is:

a. 4 b. 5 c. 6 d. 7

9. The differential equation (x+y) dx +xdy = 0 is:

a. homogeneous but not linear b. linear but not homogeneous

c. both homogeneous and linear d. neither homogeneous nor linear

10. If the equation $2x^2 + xy - y^2 - 7x - 4y + 5 = 0$ represents a pair of lines, then point of intersection is

a. (3, –2) b. (3, 1) c. (2, 1) d. (2, –1)

11. The tangents are drawn from the point (4,5) to the circle $x^2+y^2 - 4x-2y - 11 = 0$. The area of quadrilateral formed by these tangents and radii is:

a. 15sq.units b 75sq.units c. 8sq.units d. 4sq.units

12. At what point on the parabola $y^2 = 4x$; the normal makes equal angles with the co-ordinate axes:

a. (1,-2) b. (0,0) c. (4,4) d. (4,–4)

13. In an ellipse the distance between the foci is 6 and minor axis is 8, then eccentricity is:

a. $\frac{2}{5}$ b. $\frac{3}{5}$ c. $\frac{\sqrt{41}}{5}$ d. $\frac{3}{\sqrt{73}}$

14. The length of latus rectum of the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=-1$ is:

a. $\frac{8}{3}$ b. $\frac{9}{2}$ c. 0 d. 18

15. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar unit vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} + \vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ is:

a. $\frac{\pi}{4}$ b. $\frac{\pi}{2}$ c. $\frac{3\pi}{4}$ d. $\frac{\pi}{3}$

16. If in a right angled triangle ABC, the hypotenuse AB = p, then $\vec{AB}.\vec{AC} + \vec{BC}.\vec{BA} + \vec{CA}.\vec{CB}$ is :

a. $\frac{1}{2}p^2$ b. $2p^2$ c. $4p^2$ d. p^2

17. If the plane 4x+4y - kz = 6 contains the line with direction ratio 2,3,4, then k is :

a. 5 b. $\frac{7}{2}$ c. 4 d. 3

18. If $\log_a(ab) = x$, then $\log_b(ab) =$

a. $\frac{x}{1-x}$ b. $\frac{1}{x-1}$ c. $\frac{x}{x-1}$ d. $-\frac{x}{x-1}$

19. The sum of the infinite series $\frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \ldots\ldots\ldots$ is :

a. e – 2 b. e - 1 c. 1 d. 2

20. The co-efficient of x^7 in $\sum_{n=20}^{35} (1+x)^n$ is:

a. C(35,7) – C(20,7) b. C(36,7) – C(20,7)
c. C(35,7) – C(21,7) d. C(36,8) – C(20,8)

21. In how many ways 7 men and 7 women can be seated around a round table if no two women can sit together?

a. $(7!)^2$ b. 6!×6! c. 7!×6! d. 13!

22. If the roots of the equation $x^2 - 3x + 1 = 0$ are diminished by 2, then the resulting equation is:

a. $x^2+x+5 = 0$ b. $x^2 - x+5$ c. $x^2- x + 1$ d. $x^2+x-1 = 0$

23. The sum to n terms of the series 3.1.2 +3.2.3 + 3.3.4+……. is :

a. n(n+1)(n+2) b. n(n+1)(2n+1) c. (n-1)(n)(n+1) d. (n-1)(n+1)(n+2)

24. If 1, $w, w^2, w^3, \ldots\ldots\ldots, w^{n-1}$ are the nth roots of unity, then $(1 - w)(1-w^2)(1-w^3)\ldots\ldots(1-w^{n-1})$

a. 2^n b. 1 c. n d. n^2

25. If p →(qVr) is false, then the truth value of p,q,r are respectively:

a. T,F,F b. F,F,F c. F,T,F d. T,T,F

26. If the matrix A = $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ and B = $\begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$ then the system of equations $a_1x+b_1y+c_1z = d_1$; $a_2x+b_2y+c_2z = d_2$ and $a_3x+b_3y+c_3z = d_3$ has a unique solution if :

a. Rank(A) ≠ Rank(B)
b. 1≤ Rank (A), Rank(B) ≤ 3
c. Rank (A)=Rank(B) = 3
d. Rank (A) = Rank (B) = 2

27. The principal value of $\cos^{-1}(\cos\frac{5\pi}{4})$ is :

a. $\frac{5\pi}{4}$
b. $\frac{3\pi}{4}$
c. $\frac{\pi}{4}$
d. $\frac{-\pi}{4}$

28. In a binomial distribution, the mean and standard deviation are respectively 4 and $\sqrt{3}$. Then number of trials is:

a. 12
b. 36
c. 16
d. 6

29. A tower subtends angles, α, 2α, 3α respectively at points A,B and C, all lying on a horizontal line through the foot of the tower. Then AB:BC is:

a. sin3α : sin2α
b. (1+2cos2α) : 1
c. (2 + cos3α):1
d. sin2α : sin α

30. If P_1, P_2, P_3 are the altitudes of a ΔABC from the vertices A, B, C and Δ the area of Δ, then $\frac{1}{P_1^2}+\frac{1}{P_2^2}+\frac{1}{P_3^2} =$

a. $\frac{a+b+c}{\Delta}$
b. $\frac{a^2+b^2+c^2}{2\Delta^2}$
c. $\frac{a^2+b^2+c^2}{\Delta^2}$
d. $\frac{a^2+b^2+c^2}{2\Delta^2}$

…The End…

Answers: Set III

Subject: Physics

1.b	2.d	3.d	4.a	5.c	6.c	7.b	8.b	9.a	10. a
11. c	12. c	13. c	14. c	15. b	16. b	17. a	18. c	19. d	20. b
21. a	22. c	23. b	24. a	25. d	26. a	27. b	28. c	29. c	30. a

Subject: Chemistry

1.a	2.b	3.c	4.b	5.c	6.d	7.a	8.c	9.b	10. d
11. b	12. a	13. a	14. c	15. b	16. c	17. b	18. a	19. a	20. d
21. b	22. b	23. c	24. b	25. b	26. b	27. a	28. a	29. a	30. a

Subject: English

1.b	2.a	3.a	4.c	5.a	6.b	7.b	8.a	9.b	10. c
11. c	12. c	13. c	14. c	15. d	16. b	17. b	18. b	19. a	20. b
21. d	22. c	23. d	24. d	25. b	26. b	27. d	28. c	29. d	30. d

Subject: Mathematics

1.c	2.a	3.b	4.a	5.d	6.a	7.d	8.b	9.c	10. d
11. c	12. a	13. b	14. c	15. c	16. d	17. a	18. c	19. c	20. d
21. c	22. d	23. a	24. c	25. a	26. c	27. b	28. c	29. b	30. b

PEA

EDUCATION WING
EMBASSY OF INDIA
KATHMANDU, NEPAL
BE. Model Entrance Examination

Attempt all Questions. Each question carries one mark.

Choose the correct answer and blacken the appropriate bubble using HB pencil on answer sheet.

SET- IV

Subject: Physics

Time: 1 hr.

1. An aircraft is flying at a uniform speed *v* m/s. If the angle subtended at an observation point on the ground by two positions of the aircraft 't' second apart is θ, the height of the aircraft above the ground is

   a. $\frac{vt}{2\tan\theta}$ b. $\frac{2vt}{\tan\theta}$ c. $\frac{vt}{\tan\frac{\theta}{2}}$ d. $\frac{vt}{2\tan\frac{\theta}{2}}$

2. A missile is fired for maximum range from a place to another place at a distance x. The missile is detected at its half way point then

   a. Velocity of projection of missile is $\sqrt{g}\,x$

   b. Warning time is $\sqrt{\frac{x}{g}}$

   c. Speed of missile at detection is $\sqrt{gx}$

   d. Maximum height of missile is $\frac{x}{2}$

3. A machine gun is mounted on a 200 kg vehicle on a horizontal smooth road. The gun fires 10 bullets per second with a velocity of 500 m/s. The mass of each bullet is 10g then acceleration of vehicle will be

   a. 25 cm/s^2 b. 25 m/s^2 c. 50 cm/s^2 d. 50 m/s^2

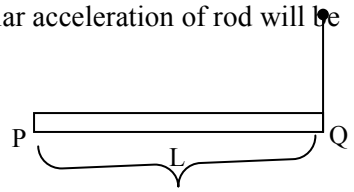
4. A body falls freely under gravity. Its velocity is *v* when it has lost potential energy equal to U. the mass of body is

   a. $\frac{U^2}{v^2}$ b. $\frac{2U^2}{v^2}$ c. $\frac{2U}{v^2}$ d. $\frac{U}{v^2}$

5. A uniform rod PQ of length L is hinged at one end and kept horizontal as shown in fig. If string is cut then initial angular acceleration of rod will be

   a. $\frac{g}{L}$ b. $\frac{2g}{3L}$

   c. $\frac{2g}{L}$ d. $\frac{3g}{2L}$



6. A small drop of water of surface tension T is squeezed between two clean glass plates so that a thin layer of thickness d and area A is formed between them. The angle of contact is zero then force required to pull the plate apart is

a. $\frac{TA}{d}$ b. $\frac{2TA}{d}$ c. $\frac{4TA}{d}$ d. $\frac{8TA}{d}$

7. The time period of body in SHM is 8 second. At t = 0, it is at the mean position, the ratio of distance travelled by it in the 1st and 2nd second is

a. $\frac{1}{2}$ b. $\frac{1}{\sqrt{2}}$ c. $\frac{1}{\sqrt{2}-1}$ d. $\frac{1}{\sqrt{3}}$

8. Tuning fork A of frequency 258 Hz give 8 beats /s with a fork B. When prong of B are cut and again A and B are sounded the number of beats heard remains same. The frequency of B is

a. 250 b. 266 c. 242 d. 258

9. The equation of motion is $y = a\cos^2\left(2\pi ft - \frac{2\pi x}{\lambda}\right)$ represent a wave with

a. Amplitude $\frac{a}{2}$ and frequency 2f b. Amplitude $\frac{a}{2}$ and frequency f

c. Amplitude a and frequency 2f d. Amplitude a and frequency f

10. A gas is at temperature 20^0c. The pressure of the gas is changed from 1.01×10^5 N/m^2 to 1.165×10^5 N/m^2. If the volume of gas is decreased isothermally by 10%. The bulk modulus is

a. 1.55×10^4 N/m^2 b. 1.01×10^5 N/m^2

c. 1.45×10^6 N/m^2 d. 1.55×10^6 N/m^2

11. When an ideal diatomic gas is heated at constant pressure the fraction of the heat energy supplied which increases the internal energy of the gas is

a. $\frac{2}{5}$ b. $\frac{3}{5}$ c. $\frac{3}{7}$ d. $\frac{5}{7}$

12. An electrical refrigerator abstract 2000 calories from ice trays. The coefficient of performance is 4 then work done by motor in calories is

a. 8000 b. 4000 c. 2000 d. 500

13. Two vessels of different materials are similar in size in every respect. The same quantity of ice is filled in them get melted in 20 minutes and 35 minutes respectively. The ratio of thermal conductivities of the metal is

a. 4 : 7 b. 7 : 4 c. 2 : 1 d. 1 : 2

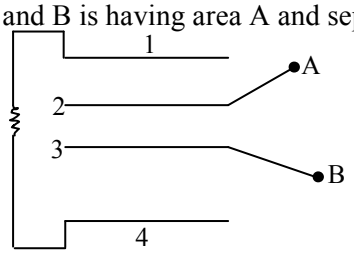
14. Two small balls each carrying a charge q are suspended by equal insulated strings of length 'l' from the hook of a stand. This arrangement is carried in a satellite in space. The tension in each string will be

a. $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4l^2}$ b. $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{l^2}$ c. $\frac{1}{4\pi\varepsilon_0}\frac{q}{4l^2}$ d. $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$

15. Charge q is placed at one of the vertex of a cube of side 'l'. The flux through one face is

a. $\frac{q}{6\varepsilon_0}$ b. $\frac{q}{12\varepsilon_0}$ c. $\frac{q}{24\varepsilon_0}$ d. $\frac{q}{48\varepsilon_0}$

16. The equivalent capacitance from fig across A and B is having area A and separation 'd' is



a. $\frac{3\varepsilon_0 A}{d}$ b. $\frac{2\varepsilon_0 A}{2}$

c. $\frac{2}{3}\frac{\varepsilon_0 A}{d}$ d. $\frac{3}{2}\frac{\varepsilon_0 A}{d}$

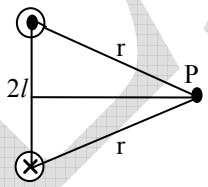
17. A wire is stretched so that its area of cross section become $\frac{1}{n}$ times, then its resistance will be

a. n times b. n^2 times c. $\frac{1}{n}$ times d. $\frac{1}{n^2}$ times

18. If two bulbs of 25w and 100w each rated at 220V are connected in series with the supply of 440V. The bulb that fuse is

a. 100w b. 25w c. None of them d .Both of them

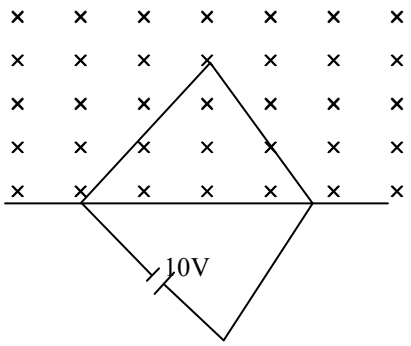
19. Equal current i is flowing in two parallel conductors distance $2l$ in mutually opposite direction as shown in fig. The magnetic induction at a point P equivalent r from both conductor is



a. $\frac{\mu_0 i l}{\pi r^2}$ b. $\frac{\mu_0 i l^2}{\pi r}$

c. $\frac{\mu_0 i l^2}{\pi r^2}$ d. $\frac{\mu_0 i l}{\pi r}$

20. A square loop of side 1m is placed in a perpendicular magnetic field. Half of the area of the loop lies inside the field. The battery of emf 10V and neglible internal resistance is connected in the loop. The magnetic field changes with time as B = (0.01 – 2t)T the total emf of battery will be



a. 1V b. 11V

c. 9V d. 10V

21. An inductance of negligible resistance whose reactance is 22Ω at 200Hz is connected to 200V 50Hz line. The inductance will be

a. 0.0175 H b. 0.175 H c. 1.75 H d. 17.5 H

22. A ray of light passes from vacuum into a medium of refractive index μ, the angle of incidence is found twice the angle of refraction. The angle of incidence is

a. $\cos^{-1}\left(\frac{\mu}{2}\right)$ b. $2\cos^{-1}\left(\frac{\mu}{2}\right)$ c. $2\sin^{-1}(\mu)$ d. $2\sin^{-1}\left(\frac{\mu}{2}\right)$

23. A biconvex lens is cut into two equal parts by a place perpendicular to the principal axis. If the power of original lens is 4D then power of each part will be

a. 2D b. 4D c. 8D d. 5D

24. In a double slit experiment, the distance between the slits is d. The screen is at a distance D from slit. If a bright fringe is found opposite to a slit on screen, the order of fringe is

a. $\frac{d^2}{2\lambda D}$ b. $\frac{2\lambda D}{d^2}$ c. $\frac{d}{\lambda D}$ d. $\frac{2d}{\lambda D}$

25. Resolving power of a telescope depends upon

a. length of telescope b. focal length of objective

c. diameter of objective d. focal length of eye lens

26. A photosensitive metal is not emitting photoelectrons when irradiated. It will do so when threshold is crossed. To cross threshold we need to increase

a. Intensity b. Frequency c. Wavelength d. None of above

27. The ratio of the areas of the circular orbits of an electrons for the 1st excited state in hydrogen atom to that for ground state is

a. 4 : 1 b. 8 : 1 c. 16 : 1 d. 32 : 1

28. Two radioactive materials x_1 and x_2 have decay constants 10λ and λ respectively. If initially they have same number of nuclei then the ratio of the number of nuclei of x_1 to that of x_2 will be $\frac{1}{e}$ after a time

a. $\frac{1}{10\lambda}$ b. $\frac{1}{11\lambda}$ c. $\frac{1}{11\lambda}$ d. $\frac{1}{9\lambda}$

29. A proton and α particle are accelerated by same potential then the wavelength will be

a. Greater for proton b. Greater for α particle

c. Equal for both d. All of above depends on condition

30. When NPN transistor is used as an amplifier, then

a. Electrons move from base to collector

b. Electrons move from to emitter

c. Electrons move from collector to base

d. Holes move from base to emitter

Subject: Chemistry

Time: 1 hr.

1. The specific heat of a metal is 0.16, its approximate atomic weight would be.
   a. 16 b. 12 c. 40 d. 20
2. The number of nodes present in 4f–orbital is.
   a. 0 b. 1 c. 2 d. 3
3. The shape of SF_4 molecule is.
   a. tetrahedral b. square planar
   c. trigonal pyramid d. see – saw
4. Which of these is not a protic acid?
   a. H_3PO_4 b. H_3PO_2 c. HPO_3 d. H_3BO_3
5. The oxidation number of chlorine in bleaching powder is.
   a. 0 b. –1 c. +1, –1 d. +1, –7
6. The packing fraction NaC*l* crystal is.
   a. 0.48 b. 0.52 c. 0.68 d. 0.74
7. Which of the following solutions would produce maximum elevation in boiling point?
   a. 0.1M glucose b. 0.2M sucrose c. 0.1M $BaCl_2$ d. 0.1M $MgSO_4$
8. k_1 and k_2 are respective equilibrium constants for the two reactions.
   i. $XeF_6(g) + H_2O(g) \rightarrow XeOF_4(g) + 2HF(g)$
   ii. $XeO_4(g) + XeF_6(g) \rightarrow eOF_4(g) + XeO_3F_2(g)$
   The equilibrium constant of the reaction $XeO_4(g) + 2HF(g) \rightarrow XeO_3F_2(g) + H_2O(g)$
   a. $\frac{k_1}{k_2^2}$ b. $k_1 - k_2$ c. $\frac{k_1}{k_2}$ d. $\frac{k_2}{k_1}$
9. The pH of a solution obtained by mixing 50m*l* of 0.4N HC*l* and 50m*l* of 0.2N NaOH is
   a. 0.1 b. 0.69 c. 2 d. 2.69
10. The rate law for a reaction between A and B is. Rate = $k[A]^m [B]^n$. On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of reaction will be.
    a. $\frac{1}{2^{m+n}}$ b. m + n c. n – m d. 2^{m-n}
11. Which of these gives maximum precipitation of As_2S_3SOl?
    a. k_2SO_4 b. $CaCl_2$ c. Na_3PO_4 d. $AlCl_3$
12. $H_2(g) + I_2(g) \longrightarrow 2HI(g)$ ΔH = 12.4k cal. According to this reaction heat of formation of HI(g) will be.
    a. 12.4k cal b. –12.4k cal c. 6.2k cal d. –6.2k cal
13. The charge required for the reduction of 1mol of MnO_4^- – ion in neutral medium is.
    a. 1F b. 3F c. 5F d. 6F
14. Which of the following has largest radius?
    a. Mg^{2+} b. Na^+ c. O^{2-} d. F^-

15. Which of the following metal reacts with H_2O at room temperature?
    a. Ag b. Fe c. Na d. A*l*
16. The sequence of ionic mobility in aqueous solution is.
    a. $k^+ > Na^+ > Rb^+ > Cs^+$ b. $Cs^+ > Rb^+ > k^+ > Na^+$
    c. $Rb^+ > k^+ > Cs^+ > Na^+$ d. $Na^+ > k^+ > Rb^+ > Cs^+$
17. The type of hybridization on B atom of diborane is.
    a. sp b. sp^2 c. sp^3 d. sp^3d
18. Sodium extract of an organic compound gives blood red colour with $FeCl_3$. It contains
    a. N b. S c. N and C*l* d. N and S
19. Which of the following compounds exhibits stereoisomerism?
    a. 3–Methyl butanoic acid b. 2–Methyl butanoic acid
    c. 2–Methyl but–1–ene d. 3–Methyl but–1–yne
20. In the sequence of reaction, Y is.

$$R-CH=CH_2 \xrightarrow{B_2H_6} X \xrightarrow{H_2O_2/OH^-} y$$

$a. R-CH_2-CHO$ $b. R-CH_2-CH_2OH$

$c.\ R-\overset{\overset{O}{\|}}{C}-CH_3$ $d. R-\underset{\underset{OH}{|}}{CH}-CH_3$

21. The process used in purification of Ti using I_2 is called.
    a. Mond's process b. Van – Arkle's process
    c. Baeyer's process d. Bosch's process
22. Which of these oxide has three electron bond?
    a. N_2O b. NO c. N_2O_3 d. N_2O_5
23. Turn bull's ppt is.
    a. $Fe_3\ [Fe(CN)_6]_2$ b. $Fe_4\ [Fe(CN)_6]_3$
    c. $k_3\ [Fe(CN)_6]$ d. $Na_4[Fe(CN)_6]$
24. The complex $[Ni(CO)_4]$ is.
    a. square planar and paramagnetic b. square planar and diamagnetic
    c. tetrahedral and paramagnetic d. tetrahedral and diagmagnetic
25. Chloroform on heating with Ag – powder gives.
    a. C_6H_6 b. C_2H_4 c. C_2H_2 d. CH_4
26. The reactive intermediate in Riemer – Timann's reaction is.
    a. Carbene b. Nitrene c. Free radical d. Carbocation
27. Acetone when heated with conc. H_2SO_4 gives rise to.
    a. Toluene b. Mesitylene c. Xylene d. Ethylbenzene
28. The reaction to convert amides into amines is name after.
    a. Perkin b. Claisen c. Hofmann d. Kekule

29. Sucrose on hydrolysis gives.
    a. glucose b. glucose and galactose
    c. glucose and fructose d. galactose and fructose
30. Natural rubber is a polymer of
    a. isoprene b. chloroprin c. butadiene d. acrylic acid

Subject: English

Time: 1 hr

Circle the right synonym (1 to 4)

1. Emulate
    a. Imitate b. Quicken c. Disgrace d. Vague
2. Lethargy
    a. Satisfied b. Greedy c. Lazy d. Gallant
3. Relinquish
    a. Maintain b. Pious c. Abandon d. Contrary
4. Chivalrous
    a. Brave b. Noble c. Cool d. Kind

Circle the right antonym (5 to 8)

5. Amicable
    a. Cordial b. Cold c. Hostile d. Repulsion
6. Blatant
    a. Boisterous b. Clamorous c. Serene d. Warm
7. Voracious
    a. Greedy b. Foil c. Gauche d. Satisfied
8. Obdurate
    a. Willing b. Noisy c. Obstinate d. Sufficiency

Circle the right choice

9. The candle was put ____ in the wind
    a. off b. down c. of d. out
10. She yearns _____ good looks
    a. for b. in c. by d. from
11. 'Let bygones be bygones' means
    a. forget the past b. to flatter
    c. to repay an old loan d. to invite trouble
12. ____China – India war was a disaster.
    a. A b. An c. The d. None
13. The fear of fire is called
    a. Pyrophobia b. Trichphobia c. Cynophobia d. Acrophobia

14. The study of different races of mankind is called
    a. Ecology b. Sociology c. Ethnology d. Graphology
15. Where _____ ?
    a. did you cut your hair b. have you cut your hair
    c. did you have your hair cut d. did you have cut your hair
16. Dr. Sita objected ____ his patients' calling at night.
    a. for b. at c. to d. against

Circle the right alternatives

17. Hammer : Carpenter
    a. Soldier : War b. Bread : Bake
    c. Salad : Rice d. Knife : Butcher
18. Flag : Nation
    a. Fox : Cunning b. Cow : Herbivorous
    c. Soldier : War d. Wine : Grapes
19. House : Room
    a. Struggle : Fight b. Boy : Girl
    c. Transport : Car d. School : College
20. Frog : Croak
    a. Hare : Levered b. Liquor : Intoxication
    c. Serpent : Hiss d. Brake : Car

Circle the right one – word

21. One who studies things of the past
    a. Historian b. Demographist c. Lapidist d. Antiguarian
22. The art of painting on a plaster surface
    a. Mosaic b. Albino c. Frescos d. Nich
23. Fear of crowds
    a. Demophobia b. Claustrophobia
    c. Ochlophobia d. Acrophobia
24. Lover of food
    a. Gourmet b. Greedy c. Stingy d. Glutton
25. A very embarrassing mistake
    a. Exgratia b. Faux pas c. Blunder d. Errata

What do the following expressions mean?

26. To smell a rat
    a. a bad smell b. suspect c. misunderstand d. hide
27. To draw a line
    a. solve b. stop c. continue d. recollect

28. To get on one's nerves

a. disapprove b. disturb c. worried d. leave

Circle the right alternative

29. She says, 'I am sorry for this.'

a. She said that she was sorry for that b. She says that she was sorry for that

c. She says that she is sorry for that d. She is saying that she is sorry for that

30. If noise is to sound then blare is to ____

a. glare b. rhythm c. pollution d. blast

Subject: Mathematics

Time: 1hr

1. The value of sum, $\sum_{n=1}^{13}(i^n + i^{n+1})$, where $i = \sqrt{-1}$, equals

a. i b. i – 1 c. – i d. 0

2. If $2^x.3^{x+4} = 7^x$, then the value of x is

a. $\frac{4\log_e 3}{\log_e 7 - \log_e 6}$ b. $\frac{4\log_e 3}{\log_e 6 - \log_e 7}$

c. $\frac{2\log_e 3}{\log_e 7 - \log_e 6}$ d. $\frac{3\log_e 4}{\log_e 6 - \log_e 7}$

3. If $A = \begin{bmatrix} \cos^2\alpha & \cos\alpha.\sin\alpha \\ \cos\alpha.\sin\alpha & \sin^2\alpha \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2\beta & \cos\beta.\sin\beta \\ \cos\beta.\sin\beta & \sin^2\beta \end{bmatrix}$ are two matrices such that the product AB is null matrix, then $\alpha - \beta$ is

a. 0 b. multiple of π

c. an odd multiple of $\frac{\pi}{2}$ d. odd multiple of π

4. The coeff. of t^{24} in the expansion of $(1 + t^2)^{12}(1 + t^{12})(1 + t^{24})$ is

a. ${}^{12}C_6 + 2$ b. ${}^{12}C_5$ c. ${}^{12}C_6$ d. ${}^{12}C_7$

5. Let T_r be the rth term of an A.P. whose first term is 'a' and common difference is 'd'. If for some positive integers m, n $m \neq n$, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then a – d equals

a. 0 b. 1 c. $\frac{1}{mn}$ d. $\frac{1}{m} + \frac{1}{n}$

6. The domain of function $f(x) = \log_2(\log_3(\log_4 x))$ is

a. $(-\infty, 4)$ b. $(4, \infty)$ c. $(0, 4)$ d. $[1, 4)$

7. If $n \in N$, and the period of $\frac{\cos nx}{\sin\left(\frac{x}{n}\right)}$ is 4π, then n is equal to

a. 4 b. 3 c. 2 d. 1

8. If $y = \log_2\log_2(x)$, then $\frac{dy}{dx}$ is equal to

a. $\frac{\log_2 e}{\log_e x}$ b. $\frac{\log_2 e}{x\log_x 2}$ c. $\frac{\log_2 x}{\log_e 2}$ d. $\frac{\log_2 e}{x\log_e x}$

9. The equation of tangent to the curve $y = 4e^{-\frac{x}{4}}$ at the point where the curve crosses y-axis is equal to

a. $3x + 4y = 16$ b. $4x + y = 4$ c. $x + y = 4$ d. $4x - 3y + 12 = 0$

10. $\int e^{x\log a} . e^x \, dx$ is equal to

a. $\frac{a^x}{\log_a e} + c$ b. $\frac{e^x}{1 + \log_e a} + c$

c. $(ae)^x + c$ d. $\frac{(ae)^x}{\log_e(ae)} + c$

11. The maximum value of the expression $12\sin\theta - 9\sin^2\theta$ is

a. 12 b. 9 c. 4 d. 3

12. If $I_n = \int_0^{\pi/4} \tan^n\theta \, d\theta$ then $I_8 + I_6$ is equal to

a. $\frac{1}{7}$ b. $\frac{1}{4}$ c. $\frac{1}{5}$ d. $\frac{1}{6}$

13. The area of the region bounded by the curves $y = e^x$, $y = \log_e x$ and lines $x = 1$, $x = 2$ is

a. $(e - 1)^2$ b. $e^2 - e + 1$

c. $e^2 - e + 1 - 2\log_e 2$ d. $(e + 1)^2$

14. The slopes of the lines represented by $x^2 + 2hxy + 2y^2 = 0$ are in the ratio 1 : 2, then h equals

a. $\pm\frac{1}{2}$ b. $\pm\frac{3}{2}$ c. ± 1 d. ± 3

15. Two tangents to the circle $x^2 + y^2 = 4$ at the points A and B meet at P(–4, 0). The area of the quadrilateral PAOB, where O is the origin is

a. 4 sq. units b. $6\sqrt{2}$ sq. units c. $4\sqrt{3}$ sq. units d. $4\sqrt{2}$ sq. units

16. If the eccentricities of the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ and the hyperbola $\frac{x^2}{64} - \frac{y^2}{b^2} = 1$ are reciprocals of each other, then b^2 is equal to

a. 191 b. 64 c. 16 d. 32

17. Suppose $\vec{a} = \lambda\vec{i} - 7\vec{j} + 3\vec{k}$, $\vec{b} = \lambda\vec{i} + \vec{j} + 2\lambda\vec{k}$. If the angle between $\vec{a}$ and $\vec{b}$ is greater than 90^0, then

a. $-7 < \lambda < 1$ b. $\lambda > 1$ c. $1 < \lambda < 7$ d. $-5 < \lambda < 1$

18. In a triangle ABC if $\angle A = 60^0$, a = 5, b = 4 then c is a root of the equation

a. $c^2 - 5c - 9 = 0$ b. $c^2 - 4c - 9 = 0$

c. $c^2 - 10c + 25 = 0$ d. $c^2 - 4c - 41 = 0$

19. The vector equation of the straight line $\frac{1-x}{3} = \frac{y+1}{-2} = \frac{3-z}{-1}$ is

a. $\vec{r} = (\vec{i} - \vec{j} + 3\vec{k}) + \lambda(3\vec{i} + 2\vec{j} - \vec{k})$ b. $\vec{r} = (\vec{i} - \vec{j} + 3\vec{k}) + \lambda(3\vec{i} - 2\vec{j} - \vec{k})$

c. $\vec{r} = (3\vec{i} - 2\vec{j} - \vec{k}) + \lambda(\vec{i} - \vec{j} + 3\vec{k})$ d. $\vec{r} = (\vec{i} - \vec{j} + 3\vec{k}) + \lambda(3\vec{i} + 2\vec{j} + \vec{k})$

20. If the angle θ between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda}\, z + 4 = 0$ is such that $\sin\theta = \frac{1}{3}$. Then, value of λ is

a. $-\frac{4}{3}$ b. $\frac{3}{4}$ c. $-\frac{3}{5}$ d. $\frac{5}{3}$

21. A house subtends a right angle at the window of an opposite house and the angle of elevation of the window from the bottom of the first house is 60^0. If the distance between the two houses be 6m, then the height of the first house is

a. $8\sqrt{3}$ m b. $6\sqrt{3}$ m c. $4\sqrt{3}$ m d. $3\sqrt{3}$

22. If three students A, B, C and solve the problem with probabilities $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$ respectively, then the probability that the problem will be solved is

a. $\frac{3}{5}$ b. $\frac{4}{5}$ c. $\frac{2}{5}$ d. $\frac{47}{60}$

23. If $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$ then, $a_1 - 2a_2 + 3a_3 - \ldots\ldots - 2na_{2n}$ is equal to

a. n b. – n c. 0 d. 2n

24. A man sells two goats at Rs. 1200 each and by doing so gain 25% on one goat and loses 25% on the other. The gain or loss percent on the whole

a. $6\frac{1}{4}$% b. $7\frac{1}{4}$% c. 25% d. no gain no loss

25. The numbers of five digits number from the integer 0, 1, 2, 3, 4, 5 which are divisible by 3 are

a. 116 b. 216 c. 128 d. 5! × 4!

26. If R be a relation defined by aRb iff $|a - b| > 0$, then the relation is

a. reflexive b. symmetric c. transitive d. symmetric and transitive

27. Two women Gangi and Jamni working separately can mow a field in 8 and 12 hours respectively. If they work for an hour alternately, Gangi beginning at 9 A.M. when will the mowing be finished?

a. 6 A.M. b. 9 A.M. c. 6:30 P.M. d. 8:30 P.M.

28. The compound interest on a certain sum for 2 years is Rs. 40.80 and the simple interest is Rs. 40. The sum and rate percent are

a. Rs. 500 and 4% b. Rs. 600 and 5%
c. Rs. 550 and 4.5% d. Rs. 700 and 4.5%

29. Number of solution of $|x - 1| = \cos x$ is

a. 0 b. 2 c. 3 d. 4

30. The value of $\cos[\tan^{-1}\{\sin(\cot^{-1}x)\}]$ is

a. $\sqrt{\frac{x^2+1}{x^2-1}}$ b. $\sqrt{\frac{1-x^2}{x^2+2}}$ c. $\sqrt{\frac{1+x^2}{2+x^2}}$ d. $\sqrt{\frac{1-x^2}{2-x^2}}$

…The End…

Answers: Set IV

Subject: Physics

1.d	2.a	3.a	4.c	5.d	6.b	7.c	8.a	9.a	10. d
11. d	12. d	13. b	14. a	15. c	16. d	17. b	18. b	19. a	20. c
21. a	22. b	23. a	24. a	25. c	26. b	27. c	28. d	29. a	30. a

Subject: Chemistry

1.c	2.d	3.d	4.d	5.c	6.d	7.c	8.d	9.c	10. d
11. d	12. c	13. b	14. c	15. c	16. b	17. c	18. d	19. b	20. b
21. b	22. b	23. a	24. d	25. c	26. a	27. c	28. c	29. c	30. a

Subject: English

1.a	2.c	3.c	4.a	5.c	6.c	7.d	8.a	9.d	10. a
11. a	12. c	13. a	14. c	15. c	16. c	17. d	18. a	19. c	20. c
21. d	22. c	23. c	24. a	25. b	26. b	27. b	28. b	29. c	30. d

Subject: Mathematics

1.b	2.a	3.c	4.a	5.a	6.b	7.c	8.d	9.c	10. d
11. c	12. a	13. c	14. b	15. a	16. a	17. a	18. b	19. a	20. d
21. a	22. a	23. b	24. a	25. b	26. d	27. c	28. a	29. b	30. c

PEA

EDUCATION WING
EMBASSY OF INDIA
KATHMANDU, NEPAL
BE. Model Entrance Examination

Attempt all Questions. Each question carries one mark.

Choose the correct answer and blacken the appropriate bubble using HB pencil on answer sheet.

SET- V

Subject: Physics

Time: 1 hr

1. A car starting from rest travels with uniform acceleration x and then comes to rest with uniform retardation y. If the total time of travel is t sec, the maximum velocity of car is

   a. $\frac{xyt}{x+y}$ b. $\frac{xy}{t(x+y)}$ c. $\frac{yt}{x(x+y)}$ d. $\frac{xt}{y(x+y)}$

2. Which one is largest when the height attained by a projectile is greatest

   a. Range b. Time of flight
   c. Angle of projection with vertical d. All of above

3. A fireman wants to slide down a rope. The breaking load for rope is $\left(\frac{3}{4}\right)^{th}$ of weight of man. The acceleration should the fireman slide down is

   a. $\frac{g}{4}$ b. $\frac{g}{2}$ c. $\frac{3g}{4}$ d. $\frac{g}{5}$

4. An object of 2kg makes an elastic collision with another object of mass M at rest and continues to move in the original direction but with one fourth of its original speed. The value of 'M' is

   a. 0.75 kg b. 1 kg c. 1.2 kg d. 6.0 kg

5. If the moment of inertia of a disc about an axis tangentially and parallel to its surface be I. Then moment of inertia about axis tangential but perpendicular to surface is

   a. $\frac{6}{5}I$ b. $\frac{3}{4}I$ c. $\frac{3}{2}I$ d. $\frac{5}{4}I$

6. The radii of two soap bubbles are r_1 and r_2 ($r_2 > r_1$) they meet to produce a double bubble. The radius of their common interface will be

   a. $\frac{r_1r_2}{r_1+r_2}$ b. $\frac{r_1+r_2}{2}$ c. $\frac{r_1r_2}{r_2-r_1}$ d. $r_2 - r_1$

7. A thin copper wire of length L increases its length by 1% when heated from temperature T_1 to T_2. The % change in area of when plate of $2L \times L$ is heated from T_1 to T_2 is

   a. 1% b. 2% c. 3% d. 4%

8. 10g of ice cubes at 0^0c are released in a tumbler of water equivalent 55g at 40^0c. The temperature of water in tumbler is

   a. 31^0c b. 22^0c c. 19^0c d. 15^0c

9. When an ideal monoatomic gas is heated at constant pressure, the fraction of heat energy supplied used to increase the internal energy of gas is

a. $\frac{2}{5}$ b. $\frac{3}{5}$ c. $\frac{3}{7}$ d. $\frac{5}{7}$

10. Two identical conducting rods are first connected independently to two vessels one containing water at 100^0c and other containing ice at 0^0c. In second case rods are placed end to end between two vessel where q_1 & q_2 be the rate of ice melt in two cases then $\frac{q_1}{q_2}$ is

a. $\frac{1}{2}$ b. $\frac{3}{1}$ c. $\frac{4}{1}$ d. $\frac{1}{4}$

11. Two waves of frequencies 256Hz and 258Hz produce beats. The interval between consecutive maxima is

a. 2 sec b. 0.5 sec c. 0.25 sec d. 1 sec

12. There are two open organ pipes of exactly the same length and material but different radii. The frequencies of their fundamental notes will be

a. wider pipe has lower frequency b. Narrower pipe has lower frequency
c. Both pipe have same frequency d. None of above

13. The intensity of sound gets reduced by 10% on passing through a slab. The reduction in intensity on passing through three consecutive slabs is

a. 30% b. 27.1% c. 20% d. 36%

14. A vessel of depth t is half filled with oil of refractive index μ_1 and other half is filled with water of refractive index μ_2. The apparent depth of vessel when viewed from above is

a. $\frac{t(\mu_1 + \mu_2)}{2\mu_1\mu_2}$ b. $\frac{t(\mu_1 - \mu_2)}{2\mu_1\mu_2}$ c. $\frac{2t(\mu_1 + \mu_2)}{\mu_1\mu_2}$ d. $\frac{2t(\mu_1 - \mu_2)}{\mu_1\mu_2}$

15. A short linear object of length L lies on the axis of a spherical mirror of focal length 'f' at a distance 'u' from the mirror. The length of image will be

a. $L\left(\frac{f}{u-f}\right)^{1/2}$ b. $L\left(\frac{u+f}{f}\right)^{1/2}$ c. $L\left(\frac{u-f}{f}\right)^{2}$ d. $L\left(\frac{f}{u-f}\right)^{2}$

16. A convex lens of power P in air is immersed in water then power of lens will be

a. increases
b. decreases
c. remain unchanged
d. increases for red colour & decreases for blue colour

17. In Young's double slit experiment, the distance between two slits is d. Interference pattern are obtained by these slits on screen at a distance D from slit. The first missing wavelength in front of one slit is

a. $\lambda = \frac{D}{d^2}$ b. $\lambda = \frac{d^2}{D}$ c. $\lambda = \frac{D}{d}$ d. $\lambda = \frac{d}{D}$

18. Three changes 4q, Q and q are placed in a straight line of length l at points at distance o, $\frac{l}{2}$ and l. Then value of Q to make net force on q zero is

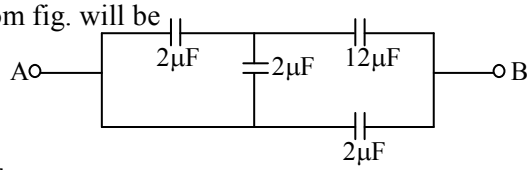
a. $-q$ b. $-2q$ c. $-\frac{q}{2}$ d. 4q

19. If a soap bubble is charged with –ve charge its radius will

a. Decreases b. Increases

c. Remain same d. May increases or decreases

20. The equivalent capacitance from fig. will be



a. $\frac{28}{9}$ μF b. 4 μF

c. 5 μF d. 18 μF

21. A wire has a resistance 10Ω. A second wire of same material is having length double and radius of cross section half of the wire. The resistance of second wire is

a. 20Ω b. 40Ω c. 80Ω d. 10Ω

22. A 12V, 24w filament bulb is supplied the current from n cells connected in series. Each cell has an emf 1.5V and internal resistance 0.25Ω. The value of n so the bulb runs at rated power is

a. 6 b. 12 c. 18 d. 24

23. The magnetic field at a point to a current carrying conductor is directly proportional to

a. Resistance of conductor b. Thickness of conductor

c. Current flowing through conductor d. Distance from conductor

24. Two magnets each of magnetic moment M are placed so as to form a right angle of cross with each other. The magnetic moment of system will be

a. 2 M b. $\sqrt{2}$ M c. $\frac{M}{2}$ d. $\frac{M}{\sqrt{2}}$

25. A coil has 200 turns and area of 70 cm^2. The magnetic field is perpendicular to the plane of coil is 0.3 wb/m^2 and rotate by 180^0 in 0.1 seconds. The value of induced emf is

a. 8.4 V b. 84 V c. 42 V d. 4.2 V

26. The maximum energy of the electrons emitted in photocell is independent of

a. Frequency of incident radiation b. Intensity of incident radiation

c. Nature of cathode d. All of these

27. The energy of electron in ground state of Hydrogen atom is –13.6 eV. The energy required to ionize the hydrogen atom in first excited state is

a. 3.4 eV b. 10.2 eV c. 13. 6 eV c. 1.9 eV

28. In a nuclear reaction

$${}_6C^{11} \longrightarrow {}_5B^{11} + {}_1e^0\ (\beta) + X$$

X is

a. proton b. neutron c. neutrino d. electron

29. The half life of a sample is 4 days. The fraction of sample will undecayed after 2 days is

a. $\sqrt{2}$ b. $\frac{1}{\sqrt{2}}$ c. $\frac{\sqrt{2}-1}{\sqrt{2}}$ d. $\frac{1}{2}$

30. The truth table belongs to

A	B	Y
1	1	0
1	0	0
0	1	0
0	0	1

a. OR b. NAND c. XOR d. NOR

Subject: Chemistry

Time: 1 hr.

1. The percentage of nitrogen in urea is about
   a. 46 b. 85 c. 18 d. 28
2. 0.24g of magnesium on reaction with an acid gave 1.42g of magnesium salt. The equivalent weight of acid is
   a. 46 b. 49 c. 60 d. 63
3. If Pauli's exclusion principle doesn't exist. The ground state electronic configuration of Li – atom is
   a. $1s^1\ 2s^2$ b. $1s^2\ 2s^1$ c. $1s^3$ d. $1s^1\ 2s^1\ 2p^1$
4. In the series of reaction
   $A \xrightarrow{-\alpha} B \xrightarrow{-2\beta} C$
   Which of these is not correct?
   a. A and B are isodiaphers b. A and C are isotopes
   c. B and C are isobars d. B and C isotones
5. Which of the following order is correct regarding their dipole moment
   a. $BF_3 > NF_3 > NH_3$ b. $NF_3 > BF_3 > NH_3$
   c. $NH_3 > NF_3 > BF_3$ d. $NH_3 > BF_3 > NF_3$
6. Which of the following species is an acid and also a conjugate base of another acid?
   a. HSO_4^- b. H_2SO_4 c. OH^- d. H_3O^+
7. Number of moles of $k_2Cr_2O_7$ reduced by 1 mole of Sn^{+2} ions is
   a. $\frac{1}{6}$ b. 6 c. $\frac{1}{3}$ d. 3
8. The indicator used in iodimetric titration is
   a. Methyl orange b. Phenolpthalein c. Starch solution d. Self indicator
9. The crystal system of a compound with unit cell dimension a = 0.387, b = 0.387, c = 0.504 nm and $\alpha = \beta = 90^0$ and $\gamma = 120^0$ is
   a. cubic c. hexagonal c. orthorhombic d. trigonal
10. Solution A contains 7 g/*l* of $MgCl_2$ and solution B contains 7 g/*l* of NaC*l*. At room temperature the osmotic pressure of
    a. A > B b. B > A c. B = A d. B ≥ A

11. Vapour density of equilibrium mixture of dissociation of PCl_5 is 70.2 at 200^0c. The degree of dissociation of PCl_5 is

a. 0.97 b. 0.242 c. 0.327 d. 0.485

12. The dissociation constant of a weak acid is 1×10^{-5}, the equilibrium constant for its reaction with strong base is

a. 1×10^{-5} b. 1×10^{9} c. 1×10^{7} d. 1×10^{14}

13. Half – life of a reaction is found to be inversely proportional to the cube of initial concentration. The order of the reaction is

a. 2 b. 3 c. 4 d. 5

14. For the adsorption of a gas on a solid, the plot of $\log \frac{x}{m}$ versus log p is linear with slope equal to

a. k b. log k c. n d. $\frac{1}{n}$

15. Which of these process is never spontaneous?

a. $\Delta H = -ve, \Delta s = -ve$ b. $\Delta H = +ve, \Delta s = +ve$

c. $\Delta H = -ve, \Delta s = +ve$ d. $\Delta H = +ve, \Delta s = -ve$

16. Which of these has the highest second ionization potential?

a. Nitrogen b. Carbon c. Oxygen d. Fluorine

17. Cyanide method used for extraction of

a. Cr b. Ag c. Cu d. Zn

18. A metal salt on heating in flame, crimson red coloured flame is obtained. The salt contains.

a. Li b. Na c. k d. Rb

19. Magnesium is present in

a. Vitamin c b. Hemoglobin c. Chlorophyll d. Vitamin B_{12}

20. Ammonium compound which on heating doesn't give NH_3 is

a. $(NH_4)_2 CO_3$ b. NH_4Cl c. $(NH_4)_2SO_4$ d. NH_4NO_2

21. Passivity of iron is due to formation of

a. FeO b. Fe_2O_3 c. Fe_3O_4 d. $FeSO_4$

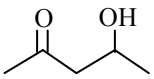
22. The type of isomerism present in nitropentamine – chromium (III) chloride is

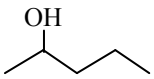
a. optical b. linkage c. ionization d. polymerization

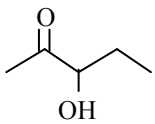
23. The number of optical isomers of tartaric acid is

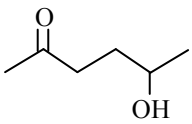
a. 2 b. 3 c. 4 d. 5

24. Which one of the following will most readily be dehydrated in acidic condition?

a. 

b. 

c. 

d. 

25. Which of these on oxidation gives ethyl methyl ketone?
    a. propanol–2 b. butanol–2 c. butanol–1 d. t–butyl alcohol
26. The reaction in which 2^0 – alcohol are oxidized to ketone by acetone in presence of aluminium t–butoxide is
    a. oppenauer oxidation b. cannizaro's reaction
    c. perkin reaction d. stephen's reduction
27. The end product C in the following sequence of reactions is
    $CH_3OH \xrightarrow{PCl_5} A \xrightarrow{KCN} B \xrightarrow{H_2O} C$
    a. HCOOH b. CH_3COOH c. CH_3NH_2 d. CH_3CONH_2
28. Hofmann Bromamide reaction is given by
    a. Amines b. esters c. alcohols d. Amides
29. Primary, secondary, tertiary amines can be separated from their mixture using
    a. Diethyl oxalate b. benzene sulphonyl chloride
    c. both d. none
30. Orlon is a polymer of
    a. Vinyl cyanide b. Vinyl Chloride
    c. Tetrafluoro ethylene d. Ethylene

Subject: English

Time: 1 hr.

Tick the right synonym (Q.No. 1-5)

1. savor
    a. taste b. quench c. swift d. embargo
2. sporadic
    a. occasional b. regular c. optimal d. spontaneous
3. undermine
    a. weary b. dirty c. weaken d. strengthen
4. verbose
    a. laconic b. vent c. suspense d. voluble
5. virile
    a. tempo b. childish c. macho d. prone

Tick the right antonym (Q. No. 6-10)

6. vibrant
    a. exciting b. lively c. dull d. lessen
7. skeptical
    a. hopeful b. doubtful c. certain d. insure

8. Melancholy

a. ecstasy b. depressed c. freedom d. happy

9. Superficial

a. profound b. mythical c. difficult d. vague

10. vivacious

a. dull b. strength c. happiness d. lively

Tick the right analogy (Q. No. 11-12)

11. crèche : Infants

a. bed linen : lunatics b. morsel : food

c. deck : sailors d. kindergarten : pupils

12. If overt is to secret in the same way as cautious is to _____

a. pompous b. rash c. clever d. coy

Tick the right one word substitute (Q. No. 13-20)

13. soldiers on horse back _____

a. infantry b. cavalry c. bayonet d. battalion

14. The study of birds

a. ornithology b. orology c. philology d. philanthropy

15. One who gives up his religion and takes another

a. postulate b. renegade c. atheist d. pantheist

16. One who studies things of the past

a. historian b. demographer c. antiquarian d. archaeologist

17. A person who has refrained from alcoholic drinks _____

a. dipsomaniac b. sober c. drunkard d. teetotaler

18. A person who displays his knowledge _____

a. erudite b. scholar c. pedant d. omniscient

19. Money paid after completion of work

a. wage b. remuneration c. dole d. honorary

20. A country that has no government _____

a. aristocracy b. mobocracy c. anarchy d. democracy

Tick the right alternative (Q. No. 21-25)

21. She <u>looked tired</u> as she had been working in the fields all morning.

a. worn down b. worn out c. worn with d. worn by

22. One mustn't _____ for class late.

a. turn down b. turn at c. turn up d. turn out

23. He refrained _____ alcohol.

a. from b. off c. against d. with

24. She walks as though she _____ modeling before

a. studies b. has studied c. had studied d. studied

25. The driver _____ to read a newspaper when he _____ for the lights to change.

a. had started, waited b. has started, has been waiting

c. was started, has waited d. started, was waiting.

Read the following passage and answer the question that follow (Q.No. 26-30)

I looked on anxiously at the clock: he was late, it was midnight and I was worried to death. What could I do? If I rang the police and then just he turned up on the doorstep I'd feel such a fool.

I knew I had to pull myself together, so I put the kettle on and made myself a cup of strong black coffee to clam my nerves. Suddenly the door bell rang and somehow I knew that something dreadful had happened. I opened the door, "Mr. Smith? I'm Constable Jones. I'm afraid there has been an accident –"

I cannot remember anything about the next few hours; it was just like a nightmare. I just could not believe that it was happening to me. Everybody was very kind, of course: they did what they could do to help and everyone said that time was a great healer. My neighbors were marvelous – they could not do enough for me. But Rover, run over by a bus! Tears kept on rolling. No other dog would ever be the same.

Questions:

26. What does the word 'he' in the first sentence refers to?

a. Mr. Smith b. Constable Jones c. Police d. Rover

27. This passage is written in ---------- style.

a. personal b. impersonal c. formal d. informal

28. Out of the following what helped the writer to bear pain?

a. strong coffee b. tears c. neighbors d. none

29. Which of the following expression can best replace, 'I was worried to death'?

a. very anxious b. very exited c. very nervous d. concerned

30. Out of the following which one is true of the writer?

a. He was worried for his neighbours

b. He was scared off police

c. He wept for his dog

d. He was waiting somebody to have coffee together.

Subject: Mathematics

Time: 1 hr

1. For all 'x', $x^2 + 2ax + 10 - 3a > 0$, then the interval in which 'a' lies is

   a. $a < -5$ b. $-5 < a < 2$ c. $a > 5$ d. $2 < a < 5$

2. If the tangents at P and Q on the parabola $y^2 = 4ax$ (whose focus is S) meet at the point T, then SP, ST and SQ are in

   a. A.P. b. G.P. c. H.P. d. Exponential formate

3. If $H_1, H_2, H_3, \ldots H_n$ be n Harmonic means between a and b then the value of $\frac{H_1 + a}{H_1 - a} + \frac{H_n + b}{H_n - b}$ is

   a. n b. 2n c. 3n d. n^2

4. What is the value of k so that the equations $x + ky + 3z = 0$, $kx + 2y + 2z = 0$ and $2x + 3y + 4z = 0$ has non-trival solution?

   a. $2, \frac{5}{4}$ b. 1, 2 c. $\frac{2}{3}, \frac{3}{4}$ d. $4, \frac{7}{2}$

5. The value of $\frac{(1+i)^8}{(1-i\sqrt{3})^3}$ is

   a. 1 b. 2 c. -2 d. ± 1

6. The ratio of circumradius and inradius of an equilateral triangle is

   a. $\frac{1}{2}$ b. $\frac{2}{3}$ c. $\frac{3}{4}$ d. $\frac{\sqrt{3}}{\sqrt{2}}$

7. For $\lim_{x \to 1} [x]$

   a. Limit exist b. Limit does not exist
   c. Continuous d. Always value 0.

8. The set of points where $f(x) = \frac{x}{1 + |x|}$ is differentiable is

   a. $(-\infty, 0) \cup (0, \infty)$ b. $(-\infty, -1) \cup (-1, \infty)$
   c. $(-\infty, \infty)$ d. $(0, \infty)$

9. If $f(x) = x^n$ then the value of

   $f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \frac{f'''(1)}{3!} + \ldots + \frac{f^n(1)}{n!}$ is

   a. 2^{-n} b. 2^n c. -2^n d. -2^{-n}

10. There are 4 balls of different colours and 4 boxes of colours same as those of the balls. Then the number of ways to place two balls in boxes with respect to their colour is

    a. 6 b. 7 c. 8 d. 12

11. On one bank of river there is a tree. On the other bank, an observer makes an angle of elevation of 60^0 at the top of tree. The angle of elevation of the top of the tree at a distance 20m away from the bank is 30^0. The width of the river is

    a. 20 m b. 10 m c. 5 m d. 1 m

12. $\lim_{n \to \infty} \left(\frac{n!}{n^n}\right)^{\frac{1}{n}} =$

    a. e b. $\frac{1}{e}$ c. $\sqrt{3}$ d. $\frac{1}{\sqrt{e}}$

13. If the sum of the coefficient in expansion of $(1 + 2x)^n$ is 6561, then the greatest term in the expansion for $x = \frac{1}{2}$ is

a. 4th term b. 5th term c. 6th term d. 8th term

14. Number of points on x-axis which are at a distance C(C < 3) from the point (2, 3) is

a. 0 b. 2 c. 3 d. 4

15. For the hyperbola $\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$, which of the following remains constant when α varies?

a. abscissae of vertices b. abscissae of foci

c. eccentricity d. directrix

16. If two circles $(x - 1)^2 + (y - 3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct point, then

a. $r > 2$ b. $2 < r < 8$ c. $r < 2$ d. $r = 2$

17. In a series of 2n observations, half of them equal 'a' and remaining half equal '–a'. If the standard deviation of the observations is 2, then |a| equals

a. $\frac{\sqrt{2}}{n}$ b. $\sqrt{2}$ c. 2 d. $\frac{1}{n}$

18. If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}$, $x > 0$ and $y = 3x$, $x > 0$, then a belongs to

a. $\left(0, \frac{1}{2}\right)$ b. $(3, \infty)$ c. $\left(\frac{1}{2}, 3\right)$ d. $\left(-3, -\frac{1}{2}\right)$

19. Range of the function $y = \sin^{-1}\left[\frac{1}{2} + x^2\right]$ where [ ] denotes the greatest integer function, is

a. $\left\{0, \frac{\pi}{2}\right\}$ b. $\left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$ c. $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$ d. $\{0, 2\pi\}$

20. The area bounded by the curve y = 2cosx and the x-axis from x = 0 to $x = 2\pi$ in square unit is

a. 4 b. 6 c. 8 d. 12

21. If $\frac{\log a}{b - c} = \frac{\log b}{c - a} = \frac{\log c}{a - b}$, then $a^a.b^b.c^c$ is equal to

a. 1 b. 2 c. – 1 d. ± 2e

22. The determinant $\begin{vmatrix} a & b & a\alpha & -b \\ b & c & b\alpha & -c \\ 2 & 1 & 0 & 0 \end{vmatrix}$ vanishes and $\alpha \neq \frac{1}{2}$, if

a. a, b, c are in A.P. b. $\alpha = \frac{1}{2}$

c. a, b, c are in G.P. d. both (a) and (b)

23. The sum of the squares of the intercepts on the axes cut off by the tangent to the curve $x^{1/3} + y^{1/3} = a^{1/3}$ (with a > 0) at $\left(\frac{a}{8}, \frac{a}{8}\right)$ is 2, then a has the value

a. 1 b. 2 c. 4 d. 8

24. If the vectors $5\vec{a} + 6\vec{b} + 7\vec{c}$, $7\vec{a} - 8\vec{b} + 9\vec{c}$ and $3\vec{a} + \lambda\vec{b} + 5\vec{c}$ are coplanar, what is the value of λ?

a. 5 b. 10 c. 15 d. 20

25. If the angle 'θ' between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \lambda z + 4 = 0$ is such that $\sin\theta = \frac{1}{3}$ then the value of λ^2 is

a. $\frac{5}{3}$ b. $-\frac{3}{5}$ c. $\frac{3}{4}$ d. $-\frac{4}{3}$

26. The line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at the point $\left(0, \frac{17}{2}, \frac{-13}{2}\right)$. Then

a. a = 2, b = 8 b. a = 4, b = 6 c. a = 6, b = 4 d. a = 8, b = 2

27. A dice is thrown 2n + 1 times, $n \in N$. The probability that faces with even numbers show odd numbers of time is

a. $\frac{2n+1}{4n+3}$ b. less than $\frac{1}{2}$ c. greater than $\frac{1}{2}$ d. $\frac{1}{2}$

28. Period of the function $f(x) = \sin\frac{\pi x}{2} + 2\cos\frac{\pi x}{3} - \tan\frac{\pi x}{4}$ is

a. 10 b. 12 c. 11 d. 9

29. If n be the number of solutions of the equation $|\cot x| = \cot x + \frac{1}{\sin x}$ $(0 < x < 2\pi)$, then value of n is

a. 2 b. 3 c. 4 d. 0

30. The value of x for which $\sin[\cot^{-1}(1 + x)] = \cos(\tan^{-1}x)$ is

a. $\frac{1}{2}$ b. 1 c. 0 d. $-\frac{1}{2}$

…The End…

Answers: Set V

Subject: Physics

1.a	2.b	3.a	4.d	5.a	6.c	7.b	8.b	9.b	10. c
11. b	12. c	13. b	14. a	15. d	16. b	17. b	18. a	19. b	20. c
21. c	22. b	23. c	24. b	25. a	26. b	27. a	28. c	29. b	30. d

Subject: Chemistry

1.a	2.c	3.c	4.d	5.c	6.a	7.c	8.c	9.b	10. b
11. d	12. b	13. c	14. b	15. d	16. c	17. b	18. a	19. c	20. d
21. c	22. b	23. b	24. a	25. b	26. a	27. b	28. d	29. c	30. a

Subject: English

1.a	2.a	3.c	4.b	5.c	6.c	7.c	8.d	9.a	10. a
11. d	12. b	13. b	14. a	15. b	16. c	17. d	18. c	19. b	20. c
21. b	22. c	23. a	24. d	25. d	26. d	27. a	28. d	29. a	30. c

Subject: Mathematics

1.b	2.b	3.b	4.a	5.c	6.a	7.b	8.c	9.b	10. a
11. b	12. b	13. b	14. a	15. b	16. b	17. c	18. c	19. a	20. c
21. a	22. d	23. c	24. d	25. a	26. c	27. d	28. b	29. a	30. d

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